3-2-2

Consider the curve defined by the equation $y+\cos y = x+1$ for $0 \le y \le 2\pi$.

1. Find d^2y/dx^2 in terms of y

Part C

2 points are earned for implicit differentiation

1 point is earned if substituted for student's dy/dx after differentiation and solves answer

$$\frac{d^2y}{dx^2} = \frac{d\left(\frac{1}{1-\sin y}\right)}{dx}$$
$$= \frac{-\left(-\cos y\frac{dy}{dx}\right)}{(1-\sin y)^2}$$
$$= \frac{\cos y\left(\frac{1}{1-\sin y}\right)}{(1-\sin y)^2}$$
$$= \frac{\cos y}{(1-\sin y)^3}$$

0	1	2	3

The student response earns three of the following points:

2 points are earned for implicit differentiation

1 point is earned if substituted for student's dy/dx after differentiation and solves answer

$$\frac{d^2y}{dx^2} = \frac{d\left(\frac{1}{1-\sin y}\right)}{dx}$$
$$= \frac{-\left(-\cos y\frac{dy}{dx}\right)}{(1-\sin y)^2}$$
$$= \frac{\cos y\left(\frac{1}{1-\sin y}\right)}{(1-\sin y)^2}$$
$$= \frac{\cos y}{(1-\sin y)^3}$$

3-2-2

2. Find dy /dx in terms of y.

Part A

2 points are earned for implicit differentiation

1 point is earned for solving $\frac{dy}{dx}$

$$\frac{dy}{dx} - \sin y \frac{dy}{dx} = 1$$
$$\frac{dy}{dx}(1 - \sin y) = 1$$
$$\frac{dy}{dx} = \frac{1}{1 - \sin y}$$

0	1	2	3

The student response earns three of the following points:

2 points are earned for implicit differentiation

1 point is earned for solving $\frac{dy}{dx}$

$$\frac{dy}{dx} - \sin y \frac{dy}{dx} = 1$$
$$\frac{dy}{dx}(1 - \sin y) = 1$$
$$\frac{dy}{dx} = \frac{1}{1 - \sin y}$$

3. Which of the following is true about the curve $x^2 - xy + y^2 = 3$ at the point $(2,\ 1)$?

(A) $\frac{dy}{dx}$ exists at $(2,\ 1)$, but there is no tangent line at that point.

(B) $\frac{dy}{dx}$ exists at $(2,\ 1)$, and the tangent line at that point is horizontal.

(C) $\frac{dy}{dx}$ exists at $(2,\ 1)$, and the tangent line at that point is neither horizontal nor vertical.

(D) $\frac{dy}{dx}$ does not exist at $(2,\ 1)$, and the tangent line at that point is vertical.

(E) $\frac{dy}{dx}$ does not exist at $(2,\ 1)$, and the tangent line at that point is horizontal.

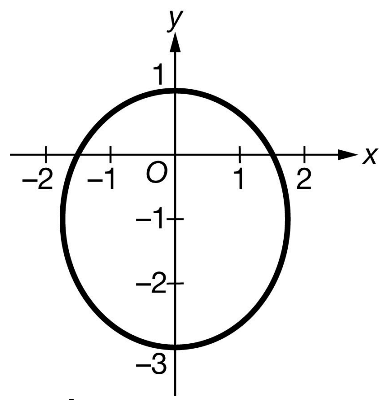
3-2-2

4. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The graph of the curve C, given by $4x^2 + 3y^2 + 6y = 9$, is shown in the figure above.

(a) Show that $\frac{dy}{dx} = \frac{-4x}{3(y+1)}$.

(b) Using the information from part (a), find $\frac{d^2y}{dx^2}$ in terms of x and y.

(c) In polar coordinates, the curve C is given by $r = \frac{3}{2+\sin\theta}$ for $0 \le \theta \le 2\pi$. Find $\frac{dr}{d\theta}$.
As θ increases, on what intervals is the distance between the origin and the point (r, θ) increasing? Give a reason for your answer.

(d) Let S be the region inside curve C, as defined in part (c), but outside the curve $r = 2$. Write, but do not evaluate, an integral expression for the area of S.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for correct implicit differentiation of both sides of the equation.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ implicit differentiation
- ☐ verification

3-2-2

Solution:

$$\frac{d}{dx}\left(4x^2+3y^2+6y\right)=\frac{d}{dx}(9)$$
$$\Rightarrow\ 8x+6y\frac{dy}{dx}+6\frac{dy}{dx}=0$$
$$\Rightarrow\ (6y+6)\frac{dy}{dx}=-8x$$
$$\Rightarrow\ \frac{dy}{dx}=\frac{-8x}{6y+6}=\frac{-4x}{3(y+1)}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for use of the quotient rule with a maximum of one error. Responses that use a rewritten form of $\frac{dy}{dx}$ in order to apply the product rule are eligible to earn this point.

The second point does not require simplification. Substitution of $\frac{-4x}{3(y+1)}$ for $\frac{dy}{dx}$ is part of the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] form of quotient rule
- [ ] $\frac{d^2y}{dx^2}$

Solution:

$$\frac{d^2y}{dx^2}=\frac{-4\cdot 3(y+1)-(-4x)\cdot 3\frac{dy}{dx}}{\left(3(y+1)\right)^2}=\frac{-12(y+1)-\frac{16x^2}{y+1}}{\left(3(y+1)\right)^2}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The second point is earned for a response that states, with or without explanation, that $\frac{dr}{d\theta}>0$; it is not necessary to state that $r>0$. The final answer may be an open or closed interval.

0	1	2

The student response accurately includes **both** of the criteria below.

3-2-2

- ☐ $\frac{dr}{d\theta}$
- ☐ answer with reason

Solution:

$r = \frac{3}{2+\sin\theta} > 0$ for $0 \leq \theta \leq 2\pi$.

$\frac{dr}{d\theta} = \frac{-3}{(2+\sin\,\theta)^2} \cdot \cos\,\theta$

$(2+\sin\,\theta)^2 > 0$ for all θ.

For $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$, $\cos\,\theta < 0$, so $\frac{dr}{d\theta} > 0$.

The distance between the origin and the point $(r,\ \theta)$ is increasing for $\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for correct limits and constants in the presence of any integrand involving $r,\ \theta$, or 2.

The second point is earned for an integrand that is the difference of squares, one of which includes θ.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ limits and constant
- ☐ form of integrand
- ☐ integrand

Solution:

$2 = \frac{3}{2+\sin\,\theta} \Rightarrow 2+\sin\,\theta = \frac{3}{2} \Rightarrow \sin\,\theta = -\frac{1}{2} \Rightarrow \theta = \frac{7\pi}{6}$ or $\theta = \frac{11\pi}{6}$

$$\text{Area} = \frac{1}{2}\int_{\frac{7\pi}{6}}^{\frac{11\pi}{6}} \left(\left(\frac{3}{2+\sin\,\theta}\right)^2 - 2^2\right) d\theta$$

3-2-2

5. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-1	0	1	2
$g(x)$	6	4	2	-1
$g'(x)$	-1	-7	-2	-3

The table above gives selected values for a differentiable and decreasing function g and its derivative. Let f be the function with $f(1)=0$ and derivative given by $f'(x)=x\sin\left(x^2\right)$.

(a) Find $f''\left(e^{1.5}\right)$. Express your answer as a decimal approximation.

(b) Let h be the function defined by $h(x)=f(g(3x))$. Find $h'(0)$. Express your answer as a decimal approximation.

(c) Write an equation for the line tangent to the graph of g^{-1}, the inverse function of g, at $x=-1$.

(d) The point $(1,4)$ lies on the curve in the xy-plane given by the equation $f(x)+(g(x))^2=xy$. What is the value of $\frac{dy}{dx}$ at the point $(1,4)$? Express your answer as a decimal approximation.

Part A

A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1 ✓

The student response accurately includes a correct decimal approximation of $f''\left(e^{1.5}\right)$

Solution:

3-2-2

$f''\left(e^{1.5}\right) = 14.144$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] chain rule (2 points for full credit)
- [ ] answer

Solution:

$h'(x) = f'(g(3x)) \cdot g'(3x) \cdot 3$

$$\begin{aligned} h'(0) &= f'(g(0)) \cdot g'(0) \cdot 3 = f'(4) \cdot (-7) \cdot 3 \\ &= 4\sin 16 \cdot (-21) = -84\sin 16 \\ &= 24.184 \text{ (or } 24.183) \end{aligned}$$

Part C

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] slope
- [ ] equation for tangent line

Solution:

Because $g(2) = -1$, $g^{-1}(-1) = 2$.

$$\left(g^{-1}\right)'(-1) = \frac{1}{g'\left(g^{-1}(-1)\right)} = \frac{1}{g'(2)} = \frac{1}{-3}$$

3-2-2

An equation for the tangent line is $y = 2 - \frac{1}{3}(x+1)$.

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the chain rule on only one side of the equation OR a correct application of the chain rule on both sides of the equation with a maximum of one computational error.

Substitution of function values is required. The third point requires a decimal approximation that is accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\frac{d}{dx}\left(f(x) + (g(x))^2\right) = \frac{d}{dx}(xy)$$

$$\Rightarrow \ f'(x) + 2g(x) \cdot g'(x) = y + x\frac{dy}{dx}$$

$$\Rightarrow \ f'(1) + 2g(1) \cdot g'(1) = 4 + \frac{dy}{dx}$$

$$\Rightarrow \ \sin 1 + 2 \cdot 2 \cdot (-2) = 4 + \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \sin 1 - 12 = -11.159 \ (\text{or } \ -11.158)$$

3-2-2

6. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	-1	0	2	3
$g(x)$	4	1	0	-2
$g'(x)$	-2	-6	-3	-4

The table above gives selected values for a differentiable and decreasing function g and its derivative. Let f be the function with $f(3) = 2$ and derivative given by $f'(x) = x\cos(e^x)$.

(a) Find $f''(\sin 2)$. Express your answer as a decimal approximation.

(b) Let h be the function defined by $h(x) = g(f(2x))$. Find $h'(1.5)$. Express your answer as a decimal approximation.

(c) Write an equation for the line tangent to the graph of g^{-1}, the inverse function of g, at $x = -2$.

(d) The point $(3, 1)$ lies on the curve in the xy-plane given by the equation $(f(x))^2 + g(x) = xy - 1$. What is the value of $\frac{dy}{dx}$ at the point $(3, 1)$? Express your answer as a decimal approximation.

Part A

A decimal approximation is required and must be accurate to three places after the decimal point.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1

The student response accurately includes a correct decimal approximation of $f''(\sin 2)$

Solution:

3-2-2

$f''(\sin 2) = -2.173 \ (\text{or } -2.172)$

Part B

The first and second points require evidence of applying the chain rule twice and no errors. At most 1 out of 3 points is earned for partial communication of chain rule with a maximum of one computational error. Substitution of function values is required. A decimal approximation is required and must be accurate to three places after the decimal point.Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- ☐ chain rule (2 points for full credit)
- ☐ answer

Solution:

$$h'(x) = g'(f(2x)) \cdot f'(2x) \cdot 2$$

$$\begin{aligned} h'(1.5) = g'(f(3)) \cdot f'(3) \cdot 2 &= g'(2) \cdot \left(3\cos\left(e^3\right)\right) \cdot 2 \\ &= -3\left(3\cos\left(e^3\right)\right) \cdot 2 \\ &= -18\cos\left(e^3\right) = -5.915 \ (\text{or } -5.914) \end{aligned}$$

Part C

The second point may be earned if response contains an incorrect value for slope based on one computational error.Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- ☐ slope
- ☐ equation for tangent line

Solution:

Because $g(3) = -2$, $g^{-1}(-2) = 3$.

$$\left(g^{-1}\right)'(-2) = \frac{1}{g'\left(g^{-1}(-2)\right)} = \frac{1}{g'(3)} = -\frac{1}{4}$$

3-2-2

An equation for the tangent line is $y = 3 - \frac{1}{4}(x+2)$.

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the chain rule on only one side of the equation OR a correct application of the chain rule on both sides of the equation with a maximum of one computational error. Substitution of function values is required. The third point requires a decimal approximation that is accurate to three places after the decimal point.Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\frac{d}{dx}\left((f(x))^2 + g(x)\right) = \frac{d}{dx}(xy-1)$$
$$\Rightarrow \ 2f(x) \cdot f'(x) + g'(x) = y + x\frac{dy}{dx}$$
$$\Rightarrow \ 2f(3) \cdot f'(3) + g'(3) = 1 + 3\frac{dy}{dx}$$
$$\Rightarrow \ 2(2) \cdot \left(3\cos\left(e^3\right)\right) + (-4) = 1 + 3\frac{dy}{dx}$$
$$\Rightarrow \frac{dy}{dx} = -0.352$$

3-2-2

7. Consider the curve defined by $x^2 + xy + y^2 = 27$.

(a) Write an expression for the slope of the curve at any point (x, y).

(b) Determine whether the lines tangent to the curve at the x-intercepts of the curve are parallel. Show the analysis that leads to your conclusion.

(c) Find the points on the curve where the lines tangent to the curve are vertical.

Part A

2 points are earned for the correct implicit differentiation

<-1> each incorrect derivative of a term

$2x + xy\prime + y + 2yy\prime = 0$

1 point is earned for correctly solving for y'

0/1 if only one y'

$$y' = \frac{-2x - y}{x + 2y}$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for the correct implicit differentiation

<-1> each incorrect derivative of a term

$2x + xy\prime + y + 2yy\prime = 0$

1 point is earned for correctly solving for y'

0/1 if only one y'

$$y' = \frac{-2x - y}{x + 2y}$$

Part B

1 point is earned for correctly using $y = 0$

3-2-2

If $y = 0, x^2 = 27$
$x = \pm 3\sqrt{3}$

1 point is earned for correctly finding y' at x-intercepts

at $x = 3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{3\sqrt{3}} = -2$

at $x = -3\sqrt{3}, y' = \frac{2 \cdot 3\sqrt{3}}{-3\sqrt{3}} = -2$

1 point is earned for the correct analysis and conclusion

Tangent lines at x-intercepts are parallel

NOTE: For solution at y-intercepts, eligible for 2nd and 3rd points

0	1	2	3

The student response earns all of the following points:

1 point is earned for correctly using $y = 0$

If $y = 0, x^2 = 27$
$x = \pm 3\sqrt{3}$

1 point is earned for correctly finding y' at x-intercepts

at $x = 3\sqrt{3}, y' = \frac{-2 \cdot 3\sqrt{3}}{3\sqrt{3}} = -2$

at $x = -3\sqrt{3}, y' = \frac{2 \cdot 3\sqrt{3}}{-3\sqrt{3}} = -2$

1 point is earned for the correct analysis and conclusion

Tangent lines at x-intercepts are parallel

NOTE: For solution at y-intercepts, eligible for 2nd and 3rd points

Part C

1 point is earned for correctly setting $x + 2y = 0$

3-2-2

y' undefined if $x + 2y = 0$

1 point is earned for correctly solving for y (or x) values

$$\begin{aligned}(-2y)^2 + (-2y)y + y^2 &= 27\\ 3y^2 &= 27\\ y &= \pm 3\end{aligned}$$

1 point is earned for correctly giving both points

Points are (-6, 3) and (6, -3)

NOTE: max 1/3 for horizontal tangents

Max 1/3 if denominator of y' is not of form $ax + by + c$ with $a \cdot b \neq 0$.

0	1	2	3

The student response earns all of the following points:

1 point is earned for correctly setting $x + 2y = 0$

y' undefined if $x + 2y = 0$

1 point is earned for correctly solving for y (or x) values

$$\begin{aligned}(-2y)^2 + (-2y)y + y^2 &= 27\\ 3y^2 &= 27\\ y &= \pm 3\end{aligned}$$

1 point is earned for correctly giving both points

Points are (-6, 3) and (6, -3)

NOTE: max 1/3 for horizontal tangents

Max 1/3 if denominator of y' is not of form $ax + by + c$ with $a \cdot b \neq 0$.

3-2-2

8. Consider the curve defined by $x^2+xy+y^2=27$.

Write an expression for the slope of the curve at any point (x, y) .

Part A

2 points are earned for the correct implicit differentiation

< -1 > each incorrect derivative of a term

$2x + xy' + y + 2yy' = 0$

1 point is earned for correctly solving for y'

0/1 if only one y'

$y' = \frac{-2x-y}{x+2y}$

0	1	2	3 ✓

The student response earns three of the following points:

2 points are earned for the correct implicit differentiation

< -1 > each incorrect derivative of a term

$2x + xy' + y + 2yy' = 0$

1 point is earned for correctly solving for y'

0/1 if only one y'

$y' = \frac{-2x-y}{x+2y}$

3-2-2

9. Consider the curve defined by $-8x^2 + 5xy + y^3 = -149$.

(a) Find $\frac{dy}{dx}$.

(b) Write an equation for the line tangent to the curve at the point $(4\ ,\ -1)$.

(c) There is a number k so that the point $(4.2\ ,\ k)$ is on the curve. Using the tangent line found in part (b), approximate the value of k .

(d) Write an equation that can be solved to find the actual value of k so that the point $(4.2\ ,\ k)$ is on the curve.

(e) Solve the equation found in part (d) for the value of k.

Part A

2 points are earned for the correct implicit differentiation

<-1> each error

$$-16x + 5y + 5x\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x - 5y}{5x + 3y^2}$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for the correct implicit differentiation

<-1> each error

$$-16x + 5y + 5x\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

Must handle at least two $\frac{dy}{dx}$ terms

3-2-2

$$\frac{dy}{dx} = \frac{16x - 5y}{5x + 3y^2}$$

Part B

1 point is earned for correctly evaluating $\frac{dy}{dx}$ at $(4, -1)$

$$\left.\frac{dy}{dx}\right|_{(4,-1)} = \frac{64+5}{20+3} = 3$$

1 point is earned for the correct equation of tangent line

$y + 1 = 3(x - 4)$

0	1	2

The student response earns all of the following points:

1 point is earned for correctly evaluating $\frac{dy}{dx}$ at $(4, -1)$

$$\left.\frac{dy}{dx}\right|_{(4,-1)} = \frac{64+5}{20+3} = 3$$

1 point is earned for the correct equation of tangent line

$y + 1 = 3(x - 4)$

Part C

1 point is earned for correctly using $x = 4.2$ in tangent line equation

$y + 1 = 3(4.2 - 4)$

1 point is earned for the correct solution

<-1> if "$k =$"

$y = -0.4$

$k \approx -0.4$

0	1	2

The student response earns all of the following points:

3-2-2

1 point is earned for correctly using $x = 4.2$ in tangent line equation

$y + 1 = 3(4.2 - 4)$

1 point is earned for the correct solution

<-1> if "k ="

$y = -0.4$

$k \approx -0.4$

Part D

1 point is earned for correctly using $x = 4.2$ in equation of curve

$$-8(4.2)^2 + 5\,(4.2)k + \ k^3 \ = -149$$

0	1 ✓

The student response earns all of the following points:

1 point is earned for correctly using $x = 4.2$ in equation of curve

$$-8(4.2)^2 + 5\,(4.2)k + \ k^3 \ = -149$$

Part E

1 point is earned for the correct answer

$k = -0.373$

0	1 ✓

The student response earns all of the following points:

1 point is earned for the correct answer

$k = -0.373$

3-2-2

10. Consider the curve defined by $-8x^2 + 5xy + y^3 = -149$.

Find $\frac{dy}{dx}$.

Part A

2 points are earned for the correct implicit differentiation

< -1 > each error

$$-16x + 5y + 5x\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x-5y}{5x+3y^2}$$

0	1	2	3

The student response earns three of the following points:

2 points are earned for the correct implicit differentiation

< -1 > each error

$$-16x + 5y + 5x\frac{dy}{dx} + 3y^2\frac{dy}{dx} = 0$$

1 point is earned for correctly solving for $\frac{dy}{dx}$

Must handle at least two $\frac{dy}{dx}$ terms

$$\frac{dy}{dx} = \frac{16x-5y}{5x+3y^2}$$

3-2-2

11. Consider the curve given by $x^2 + 4y^2 = 7 + 3xy$.

(a) Show that $\frac{dy}{dx} = \frac{3y-2x}{8y-3x}$.

(b) Show that there is a point P with x-coordinate 3 at which the line tangent to the curve at P is horizontal. Find the y-coordinate of P.

(c) Find the value of $\frac{d^2y}{dx^2}$ at the point P found in part (b). Does the curve have a local maximum, a local minimum, or neither at the point P? Justify your answer.

Part A

1 point is earned for implicit differentiation

1 point is earned for solving for y'

$$\begin{aligned} 2x + 8yy' &= 3y + 3xy' \\ (8y - 3x)y' &= 3y - 2x \\ y' &= \frac{3y-2x}{8y-3x} \end{aligned}$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for implicit differentiation

1 point is earned for solving for y'

$$\begin{aligned} 2x + 8yy' &= 3y + 3xy' \\ (8y - 3x)y' &= 3y - 2x \\ y' &= \frac{3y-2x}{8y-3x} \end{aligned}$$

Part B

1 point is earned for: $\frac{dy}{dx} = 0$

1 point is earned for showing slope is 0 at (3, 2)

1 point is earned for showing (3, 2) lies on curve

$$\frac{3y-2x}{8y-3x} = 0;\ 3y - 2x = 0$$

3-2-2

When $x = 3$, $3y = 6$

$y = 2$

$3^2 + 4 \cdot 2^2 = 25$ and $7 + 3 \cdot 3 \cdot 2 = 25$

Therefore, $P = (3, 2)$ is on the curve and the slope is 0 at this point.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for: $\frac{dy}{dx} = 0$

1 point is earned for showing slope is 0 at (3, 2)

1 point is earned for showing (3, 2) lies on curve

$$\frac{3y - 2x}{8y - 3x} = 0;\ 3y - 2x = 0$$

When $x = 3$, $3y = 6$

$y = 2$

$3^2 + 4 \cdot 2^2 = 25$ and $7 + 3 \cdot 3 \cdot 2 = 25$

Therefore, $P = (3, 2)$ is on the curve and the slope is 0 at this point.

Part C

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for the value of $\frac{d^2y}{dx^2}$ at (3, 2)

1 point is earned for the conclusion with justification

$$\frac{d^2y}{dx^2} = \frac{(8y-3x)(3y'-2)-(3y-2x)(8y'-3)}{(8y-3x)^2}$$

At $P = (3, 2)$, $\frac{d^2y}{dx^2} = \frac{(16-9)(-2)}{(16-9)^2} = -\frac{2}{7}$.

Since $y' = 0$ and $y'' < 0$ at P, the curve has a local maximum at P.

3-2-2

0	1	2	3	4

The student response earns all of the following points:

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for the value of $\frac{d^2y}{dx^2}$ at (3, 2)

1 point is earned for the conclusion with justification

$$\frac{d^2y}{dx^2} = \frac{(8y-3x)(3y'-2)-(3y-2x)(8y'-3)}{(8y-3x)^2}$$

At $P = (3, 2)$, $\frac{d^2y}{dx^2} = \frac{(16-9)(-2)}{(16-9)^2} = -\frac{2}{7}$.

Since $y' = 0$ and $y'' < 0$ at P, the curve has a local maximum at P.

3-2-2

Consider the curve given by $x^2 + 4y^2 = 7 + 3xy$.

12. Show that $\frac{dy}{dx} = \frac{3y-2x}{8y-3x}$.

Part A

1 point is earned for implicit differentiation

1 point is earned for solves for y'

$2x + 8yy' = 3y + 3xy'$
$(8y - 3x)y' = 3y - 2x$
$y' = \frac{3y-2x}{8y-3x}$

0	1	2

The student response earns all of the following points:

1 point is earned for implicit differentiation

1 point is earned for solves for y'

$2x + 8yy' = 3y + 3xy'$
$(8y - 3x)y' = 3y - 2x$
$y' = \frac{3y-2x}{8y-3x}$

3-2-2

13. Find the value of $\frac{d^2y}{dx^2}$ at the point P found in part (b). Does the curve have a local maximum, a local minimum, or neither at the point P? Justify your answer.

Part C

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for the value of $\frac{d^2y}{dx^2}$ at (3, 2)

1 point is earned for the conclusion with justification

$$\frac{d^2y}{dx^2} = \frac{(8y-3x)(3y'-2)-(3y-2x)(8y'-3)}{(8y-3x)^2}$$

At $P = (3, 2)$, $\frac{d^2y}{dx^2} = \frac{(16-9)(-2)}{(16-9)^2} = -\frac{2}{7}$.

Since $y' = 0$ and $y'' < 0$ at P, the curve has a local maximum at P.

0	1	2	3	4

The student response earns all of the following points:

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for the value of $\frac{d^2y}{dx^2}$ at (3, 2)

1 point is earned for the conclusion with justification

$$\frac{d^2y}{dx^2} = \frac{(8y-3x)(3y'-2)-(3y-2x)(8y'-3)}{(8y-3x)^2}$$

At $P = (3, 2)$, $\frac{d^2y}{dx^2} = \frac{(16-9)(-2)}{(16-9)^2} = -\frac{2}{7}$.

Since $y' = 0$ and $y'' < 0$ at P, the curve has a local maximum at P.

3-2-2

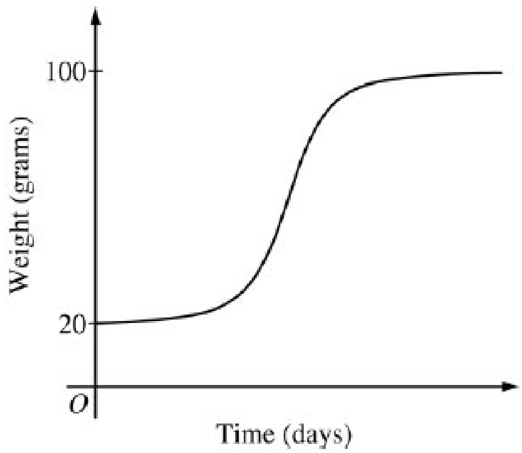
14. The rate at which a baby bird gains weight is proportional to the difference between its adult weight and its current weight. At time $t = 0$, when the bird is first weighed, its weight is 20 grams. If $B(t)$ is the weight of the bird, in grams, at time t days after it is first weighed, then

$$\frac{dB}{dt} = \frac{1}{5}(100 - B).$$

Let $y = B(t)$ be the solution to the differential equation above with initial condition $B(0) = 20$.

(a) Is the bird gaining weight faster when it weighs 40 grams or when it weighs 70 grams? Explain your reasoning.

(b) Find $\frac{d^2B}{dt^2}$ in terms of B. Use $\frac{d^2B}{dt^2}$ to explain why the graph of B cannot resemble the following graph.



(c) Use separation of variables to find $y = B(t)$, the particular solution to the differential equation with initial condition $B(0) = 20$.

Part A

1 point is earned for using $\frac{dB}{dt}$

1 point is earned for the answer with reason

$$\left.\frac{dB}{dt}\right|_{B=40} = \frac{1}{5}(60) = 12$$
$$\left.\frac{dB}{dt}\right|_{B=70} = \frac{1}{5}(30) = 6$$

Because $\left.\frac{dB}{dt}\right|_{B=40} > \left.\frac{dB}{dt}\right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for using $\frac{dB}{dt}$

3-2-2

1 point is earned for the answer with reason

$$\left.\frac{dB}{dt}\right|_{B=40} = \tfrac{1}{5}(60) = 12$$
$$\left.\frac{dB}{dt}\right|_{B=70} = \tfrac{1}{5}(30) = 6$$

Because $\left.\frac{dB}{dt}\right|_{B=40} > \left.\frac{dB}{dt}\right|_{B=70}$, the bird is gaining weight faster when it weighs 40 grams.

Part B

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5}\frac{dB}{dt} = -\frac{1}{5}\cdot\frac{1}{5}(100 - B) = -\frac{1}{25}(100 - B)$$

Therefore, the graph of B is concave down for $20 \le B < 100$. A portion of the given graph is concave up.

0	1	2

The student earns all of the following points:

1 point is earned for the $\frac{d^2B}{dt^2}$ in terms of B

1 point is earned for the explanation

$$\frac{d^2B}{dt^2} = -\frac{1}{5}\frac{dB}{dt} = -\frac{1}{5}\cdot\frac{1}{5}(100 - B) = -\frac{1}{25}(100 - B)$$

Therefore, the graph of B is concave down for $20 \le B < 100$. A portion of the given graph is concave up.

Part C

1 point is earned for the separation of variables

3-2-2

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

$\frac{dB}{dt} = \frac{1}{5}(100 - B)$
$\int \frac{1}{100-B} dB = \int \frac{1}{5} dt$
$-\ln|100 - B| = \frac{1}{5}t + c$
Because $20 \leq B < 100,\ \ |100 - B| = 100 - B.$
$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$
$100 - B = 80e^{-t/5}$
$B(t) = 100 - 80e^{-t/5},\ t \geq 0$

0	1	2	3	4	5

The student earns all of the following points:

1 point is earned for the separation of variables

1 point is earned for the antiderivatives

1 point is earned for the constant of integration

1 point is earned for using initial condition

1 point is earned for solving for B

$\frac{dB}{dt} = \frac{1}{5}(100 - B)$
$\int \frac{1}{100-B} dB = \int \frac{1}{5} dt$
$-\ln|100 - B| = \frac{1}{5}t + c$
Because $20 \leq B < 100,\ \ |100 - B| = 100 - B.$
$-\ln(100 - 20) = \frac{1}{5}(0) + C \Rightarrow -\ln(80) = C$
$100 - B = 80e^{-t/5}$
$B(t) = 100 - 80e^{-t/5},\ t \geq 0$

3-2-2

15. Which of the following is an equation of the line tangent to the graph of $x^2 - 3xy = 10$ at the point $(1, -3)$?

(A) $y + 3 = -11\,(x - 1)$

(B) $y + 3 = -\frac{7}{3}(x - 1)$

(C) $y + 3 = \frac{1}{3}(x - 1)$

(D) $y + 3 = \frac{7}{3}(x - 1)$

(E) $y + 3 = \frac{11}{3}(x - 1)$

3-2-2

16. Consider the curve defined by the equation $y + \cos y = x + 1$ for $0 \leq y \leq 2\pi$.

(a) Find dy/dx in terms of y.

(b) Write an equation for each vertical tangent to the curve

(c) Find d^2y/dx^2 in terms of y.

Part A

2 points are earned for implicit differentiation

1 point is earned for solving dy/dx

$$\frac{dy}{dx} - \sin y \frac{dy}{dx} = 1$$
$$\frac{dy}{dx}(1 - \sin y) = 1$$
$$\frac{dy}{dx} = \frac{1}{1 - \sin y}$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for implicit differentiation

1 point is earned for solving dy/dx

$$\frac{dy}{dx} - \sin y \frac{dy}{dx} = 1$$
$$\frac{dy}{dx}(1 - \sin y) = 1$$
$$\frac{dy}{dx} = \frac{1}{1 - \sin y}$$

Part B

1 point is earned for finding y where dy/dx does not exist

1 point is earned for solving that equation for y

1 point is earned for using that y solution to give equation of vertical line

dy/dx undefined when $\sin y = 1$

$y = \pi/2$

$\pi/2 + 0 = x + 1$

3-2-2

$x = \pi/2 - 1$

0	1	2	3

The student response earns all of the following points:

1 point is earned for finding y where dy/dx does not exist

1 point is earned for solving that equation for y

1 point is earned for using that y solution to give equation of vertical line

dy/dx undefined when $\sin y = 1$

$y = \pi/2$

$\pi/2 + 0 = x + 1$

$x = \pi/2 - 1$

Part C

2 points are earned for implicit differentiation

1 point is earned if substituted for student's dy/dx after differentiation and solves answer $\frac{d^2y}{dx^2}$

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d\left(\frac{1}{1-\sin y}\right)}{dx} \\ &= \frac{-\left(-\cos y\frac{dy}{dx}\right)}{(1-\sin y)^2} \\ &= \frac{\cos y\frac{1}{1-\sin y}}{(1-\sin y)^2} \\ &= \frac{\cos y}{(1-\sin y)^3}\end{aligned}$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for implicit differentiation

3-2-2

1 point is earned if substituted for student's dy/dx after differentiation and solves answer $\frac{d^2y}{dx^2}$

$$\frac{d^2y}{dx^2} = \frac{d\left(\frac{1}{1-\sin y}\right)}{dx}$$
$$= \frac{-\left(-\cos y\frac{dy}{dx}\right)}{(1-\sin y)^2}$$
$$= \frac{\cos y\frac{1}{1-\sin y}}{(1-\sin y)^2}$$
$$= \frac{\cos y}{(1-\sin y)^3}$$

17. If $3x^2 + 2xy + y^2 = 1$, then $\frac{dy}{dx} =$

(A) $-\frac{3x+y}{y^2}$

(B) $-\frac{3x+y}{x+y}$ ✓

(C) $\frac{1-3x-y}{x+y}$

(D) $-\frac{3x}{1+y}$

(E) $-\frac{3x}{x+y}$

18. If $\tan(xy)=x$, then $\frac{dy}{dx} =$

(A) $\frac{1-y\tan(xy)\sec(xy)}{x\tan(xy)\sec(xy)}$

(B) $\frac{\sec^2(xy)-y}{x}$

(C) $\cos^2(xy)$

(D) $\frac{\cos^2(xy)}{x}$

(E) $\frac{\cos^2(xy)-y}{x}$ ✓

3-2-2

19. Consider the differential equation $\frac{dy}{dx} = \frac{1}{2}x + y - 1$

Find d^2y/dx^2 in terms of x and y. Describe the region in the xy-plane in which all solution curves to the differential equation are concave up.

Part B

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$$

Solution curves will be concave up on the half-plane above the line

$$y = -\frac{1}{2}x + \frac{1}{2}.$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$$

Solution curves will be concave up on the half-plane above the line

$$y = -\frac{1}{2}x + \frac{1}{2}.$$

3-2-2

20. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Consider the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$.

(a) Show that $\frac{d^2y}{dx^2} = \frac{2y^3 - y^2 - 2xy}{x^2}$.

(b) Let $y = g(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$ with initial condition $g(4) = 2$. Does g have a relative minimum, a relative maximum, or neither at $x = 4$? Justify your answer.

(c) Let $y = h(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = -1 + \frac{y^2}{x}$ with initial condition $h(1) = 2$. Write the second-degree Taylor polynomial for h about $x = 1$.

(d) For the function h given in part (c), it is known that $|h'''(x)| \le 60$ for all x in the interval $0.9 \le x \le 1.1$. Let A represent the approximation of $h(1.1)$ found by using the second-degree Taylor polynomial for h about $x = 1$ from part (c). Use the Lagrange error bound to show that A differs from $h(1.1)$ by at most 0.01.

Part A

1 point is earned for: quotient rule

1 point is earned for: $\frac{d^2y}{dx^2}$

$$\frac{d^2y}{dx^2} = \frac{x \cdot 2y\frac{dy}{dx} - y^2 \cdot 1}{x^2}$$
$$= \frac{2xy\left(-1 + \frac{y^2}{x}\right) - y^2}{x^2} = \frac{2y^3 - y^2 - 2xy}{x^2}$$

0	1	2

The student response earns all of the following points:

1 point is earned for: quotient rule

1 point is earned for: $\frac{d^2y}{dx^2}$

$$\frac{d^2y}{dx^2} = \frac{x \cdot 2y\frac{dy}{dx} - y^2 \cdot 1}{x^2}$$
$$= \frac{2xy\left(-1 + \frac{y^2}{x}\right) - y^2}{x^2} = \frac{2y^3 - y^2 - 2xy}{x^2}$$

Part B

1 point is earned for: considers $\left.\frac{dy}{dx}\right|_{(x,y)=(4,\,2)}$

3-2-2

1 point is earned for: answer with justification

$$\left.\frac{dy}{dx}\right|_{(x,y)=(4,\,2)} = -1 + \frac{4}{4} = 0$$
$$\left.\frac{d^2y}{dx^2}\right|_{(x,y)=(4,\,2)} = \frac{2\cdot 8 - 4 - 16}{16} = -\frac{1}{4} < 0$$

By the Second Derivative Test, g has a relative maximum at $x = 4$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: considers $\left.\frac{dy}{dx}\right|_{(x,y)=(4,\,2)}$

1 point is earned for: answer with justification

$$\left.\frac{dy}{dx}\right|_{(x,y)=(4,\,2)} = -1 + \frac{4}{4} = 0$$
$$\left.\frac{d^2y}{dx^2}\right|_{(x,y)=(4,\,2)} = \frac{2\cdot 8 - 4 - 16}{16} = -\frac{1}{4} < 0$$

By the Second Derivative Test, g has a relative maximum at $x = 4$.

Part C

1 point is earned for: uses $\left.\frac{dy}{dx}\right|_{(x,\,y)=(1,\,2)}$ and $\left.\frac{d^2y}{dx^2}\right|_{(x,\,y)=(1,\,2)}$

2 points are earned for: Taylor polynomial

$$\left.\frac{dy}{dx}\right|_{(x,y)=(1,\,2)} = -1 + \frac{4}{1} = 3$$
$$\left.\frac{d^2y}{dx^2}\right|_{(x,y)=(1,\,2)} = \frac{2\cdot 8 - 4 - 4}{1} = 8$$

The second-degree Taylor polynomial for h about $x = 1$ is
$T_2(x) = 2 + 3(x-1) + \frac{8}{2!}(x-1)^2 = 2 + 3(x-1) + 4(x-1)^2$.

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned for: uses $\left.\frac{dy}{dx}\right|_{(x,\,y)=(1,\,2)}$ and $\left.\frac{d^2y}{dx^2}\right|_{(x,\,y)=(1,\,2)}$

3-2-2

2 points are earned for: Taylor polynomial

$$\left.\frac{dy}{dx}\right|_{(x,y)=(1,\,2)} = -1 + \frac{4}{1} = 3$$
$$\left.\frac{d^2y}{dx^2}\right|_{(x,y)=(1,\,2)} = \frac{2\cdot 8-4-4}{1} = 8$$

The second-degree Taylor polynomial for h about $x = 1$ is
$T_2\left(x\right) = 2 + 3\left(x-1\right) + \frac{8}{2!}(x-1)^2 = 2 + 3\left(x-1\right) + 4(x-1)^2$.

Part D

1 point is earned for: form of the error bound

1 point is earned for: analysis

$$\left|h\left(1.1\right) - A\right| \le \frac{\max\limits_{1.0\le x\le 1.1}\left|h'''(x)\right|\left|1.1-1\right|^3}{3!} \le \frac{60}{6}\cdot\frac{1}{1000} = \frac{10}{1000} = \frac{1}{100}$$

0	1	2

The student response earns all of the following points:

1 point is earned for: form of the error bound

1 point is earned for: analysis

$$\left|h\left(1.1\right) - A\right| \le \frac{\max\limits_{1.0\le x\le 1.1}\left|h'''(x)\right|\left|1.1-1\right|^3}{3!} \le \frac{60}{6}\cdot\frac{1}{1000} = \frac{10}{1000} = \frac{1}{100}$$

3-2-2

21. The function f is differentiable for all real numbers. The point $\left(3, \frac{1}{4}\right)$ is on the graph of $y = f(x)$, and the slope at each point (x, y) on the graph is given by $\frac{dy}{dx} = y^2\,(6 - 2x)$.

(a) Find $\frac{d^2y}{dx^2}$ and evaluate it at the point $\left(3, \frac{1}{4}\right)$.

(b) Find $y = f(x)$ by solving the differential equation $\frac{dy}{dx} = y^2\,(6 - 2x)$ with the initial condition $f(3) = 1/4$.

Part A

2 points are earned for $\frac{d^2y}{dx^2}$

<-2> product rule or chain rule error

1 point is earned for the value at $\left(3, \frac{1}{4}\right)$

$$
\begin{aligned}
\frac{d^2y}{dx^2} &= 2y\frac{dy}{dx}(6 - 2x) - 2y^2 \\
&= 2y^3(6 - 2x)^2 - 2y^2 \\
\left.\frac{d^2y}{dx^2}\right|_{\left(3, \frac{1}{4}\right)} &= 0 - 2\left(\tfrac{1}{4}\right)^2 = -\tfrac{1}{8}
\end{aligned}
$$

0	1	2	3

The student response earns all of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

<-2> product rule or chain rule error

1 point is earned for the value at $\left(3, \frac{1}{4}\right)$

$$
\begin{aligned}
\frac{d^2y}{dx^2} &= 2y\frac{dy}{dx}(6 - 2x) - 2y^2 \\
&= 2y^3(6 - 2x)^2 - 2y^2 \\
\left.\frac{d^2y}{dx^2}\right|_{\left(3, \frac{1}{4}\right)} &= 0 - 2\left(\tfrac{1}{4}\right)^2 = -\tfrac{1}{8}
\end{aligned}
$$

Part B

1 point is earned for the separates variables

3-2-2

1 point is earned for the antiderivative of *dy* term

1 point is earned for the antiderivative of *dx* term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $f(3) = 1/4$

1 point is earned for solves for *y*

$\frac{1}{y^2}dy = (6 - 2x)dx$

$-\frac{1}{y} = 6x - x^2 + C$

$-4 = 18 - 9 + C = 9 + C$

$C = -13$

$y = \frac{1}{x^2 - 6x + 13}$

0	1	2	3	4	5	6

The student response earns all of the following points:

1 point is earned for the separates variables

1 point is earned for the antiderivative of *dy* term

1 point is earned for the antiderivative of *dx* term

1 point is earned for the constant of integration

1 point is earned for the uses initial condition $f(3) = 1/4$

1 point is earned for solves for *y*

$\frac{1}{y^2}dy = (6 - 2x)dx$

$-\frac{1}{y} = 6x - x^2 + C$

$-4 = 18 - 9 + C = 9 + C$

$C = -13$

$y = \frac{1}{x^2 - 6x + 13}$

3-2-2

22. The function f is differentiable for all real numbers. The point $\left(3, \frac{1}{4}\right)$ is on the graph of $y = f(x)$, and the slope at each point (x, y) on the graph is given by $\frac{dy}{dx} = y^2(6 - 2x)$.

Find $\frac{d^2y}{dx^2}$ and evaluate it at the point $\left(3, \frac{1}{4}\right)$.

Part A

2 points are earned for $\frac{d^2y}{dx^2}$

< -2 > product rule or chain rule error

1 point is earned for the value at $\left(3, \frac{1}{4}\right)$

$$\frac{d^2y}{dx^2} = 2y\frac{dy}{dx}(6 - 2x) - 2y^2$$
$$= 2y^3(6 - 2x)^2 - 2y^2$$
$$\left.\frac{d^2y}{dx^2}\right|_{\left(3, \frac{1}{4}\right)} = 0 - 2\left(\frac{1}{4}\right)^2 = -\frac{1}{8}$$

0	1	2	3

The student response earns three of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

< -2 > product rule or chain rule error

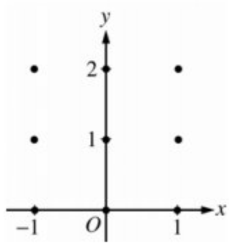
1 point is earned for the value at $\left(3, \frac{1}{4}\right)$

$$\frac{d^2y}{dx^2} = 2y\frac{dy}{dx}(6 - 2x) - 2y^2$$
$$= 2y^3(6 - 2x)^2 - 2y^2$$
$$\left.\frac{d^2y}{dx^2}\right|_{\left(3, \frac{1}{4}\right)} = 0 - 2\left(\frac{1}{4}\right)^2 = -\frac{1}{8}$$

3-2-2

23. Consider the differential equation $\frac{dy}{dx} = \frac{1}{2}x + y - 1$.

(a) On the axes provided, sketch a slope field for the given differential equation at the nine points indicated.



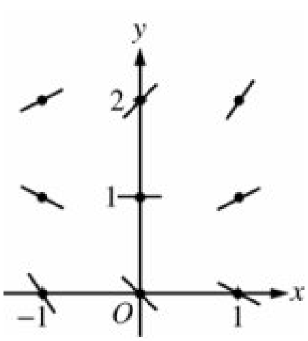
(b) Find $\frac{d^2y}{dx^2}$ in terms of x and y. Describe the region in the xy-plane in which all solution curves to the differential equation are concave up.

(c) Let $y = f(x)$ be a particular solution to the differential equation with the initial condition $f(0) = 1$. Does f have a relative minimum, a relative maximum, or neither at $x = 0$? Justify your answer.

(d) Find the values of the constants m and b, for which $y = mx + b$ is a solution to the differential equation.

Part A

2 points are earned for sign of each slope at each point and relative steepness of slope lines in rows and columns

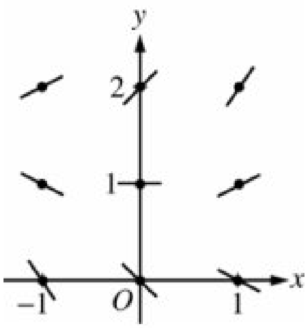


0	1	2

The student response earns all of the following points:

2 points are earned for sign of each slope at each point and relative steepness of slope lines in rows and columns

3-2-2



Part B

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$

Solution curves will be concave up on the half-plane above the line

$y = -\frac{1}{2}x + \frac{1}{2}$.

0	1	2	3

The student response earns all of the following points:

2 points are earned for $\frac{d^2y}{dx^2}$

1 point is earned for the description

$\frac{d^2y}{dx^2} = \frac{1}{2} + \frac{dy}{dx} = \frac{1}{2}x + y - \frac{1}{2}$

Solution curves will be concave up on the half-plane above the line

$y = -\frac{1}{2}x + \frac{1}{2}$.

Part C

1 point is earned for the answer

1 point is earned for the justification

$\left.\frac{dy}{dx}\right|_{(0,1)} = 0 + 1 - 1 = 0$ and $\left.\frac{d^2y}{dx^2}\right|_{(0,1)} = 0 + 1 - \frac{1}{2} > 0$

3-2-2

Thus, f has a relative minimum at $(0,1)$.

0	1	2

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for the justification

$$\left.\frac{dy}{dx}\right|_{(0,1)} = 0 + 1 - 1 = 0 \text{ and } \left.\frac{d^2y}{dx^2}\right|_{(0,1)} = 0 + 1 - \frac{1}{2} > 0$$

Thus, f has a relative minimum at $(0,1)$.

Part D

1 point is earned for the value of m

1 point is earned for the value of b

Substituting $y = mx + b$ into the differential equation:

$$m = \frac{1}{2}x + (mx + b) - 1 = (m + \frac{1}{2})x + (b - 1)$$

Then $0 = m + \frac{1}{2}$ and $m = b - 1$: $m = -\frac{1}{2}$ and $b = \frac{1}{2}$.

0	1	2

The student response earns all of the following points:

1 point is earned for the value of m

1 point is earned for the value of b

Substituting $y = mx + b$ into the differential equation:

$$m = \frac{1}{2}x + (mx + b) - 1 = (m + \frac{1}{2})x + (b - 1)$$

Then $0 = m + \frac{1}{2}$ and $m = b - 1$: $m = -\frac{1}{2}$ and $b = \frac{1}{2}$.

3-2-2

24. Consider the curve given by the equation $x^3 - 2xy + y = 22$. It can be shown that $\frac{dy}{dx} = \frac{2y-3x^2}{1-2x}$.

Part A

Find the slope of the line tangent to the curve at the point $(1, -21)$. Show the work that leads to your answer.

Part B

Find the coordinates of all points on the curve at which the line tangent to the curve is vertical, or explain why no such points exist.

Part C

The line tangent to the curve at the point $(-2, 6)$ is horizontal. Use the Second Derivative Test to determine whether the curve has a relative minimum or a relative maximum at $(-2, 6)$. Show the work that leads to your answer.

Part D

For the curve defined by the equation $y + x\cos y = 18$, find $\frac{dy}{dx}$. Show the work that leads to your answer.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response of $\frac{2(-21)-3(1)^2}{1-2(1)}$ banks **P1** (i.e., subsequent errors in simplification will not be considered in scoring for **P1**).
- A response of $\frac{2(-21)-3}{1-2}$ is sufficient to earn **P1**.
- A response of 45 with no supporting work does not earn **P1**.

Model Solution:

At $(1, -21)$, $\frac{dy}{dx} = \frac{2(-21)-3(1)^2}{1-2(1)} = 45$.

The slope of the tangent line is 45.

0	1

The response accurately includes the criteria below.

- Answer with supporting work

3-2-2

Part B Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response. Scoring Notes:

- **P2** can be earned with a response of $1-2x=0$, $1=2x$, or $x=\frac{1}{2}$.

Model Solution:

A vertical tangent line can only occur when $1-2x=0$ or when $x=\frac{1}{2}$.

0	1 ✓

The response accurately includes the criteria below.

- Considers $1-2x=0$

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To be eligible for **P3**, a response must have earned **P2**.
- To earn **P3**, a response must provide evidence that no points with $x=\frac{1}{2}$ lie on the curve.

Model Solution:

For all points (x,y) on the curve, $x^3-2xy+y=22$.

However, when $x=\frac{1}{2}$, $\left(\frac{1}{2}\right)^3-2\cdot\frac{1}{2}\cdot y+y=\frac{1}{8}-y+y=\frac{1}{8}\neq 22$.

Therefore, there are no points on the curve with $x=\frac{1}{2}$. Thus, there are no points on the curve with a vertical tangent line.

0	1 ✓

The response accurately includes the criteria below.

- Answer with explanation

Part C Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

3-2-2

- To earn **P4**, at least four of the five factors in an application of the quotient rule must be correct. The five correct factors are $1-2x$, $2\frac{dy}{dx}-6x$, $2y-3x^2$, -2, and $(1-2x)^2$.
- Special cases: The following examples earn **P4**.
  - $\frac{d^2y}{dx^2}=\frac{(2y-3x^2)(-2)-(1-2x)\left(2\frac{dy}{dx}-6x\right)}{(1-2x)^2}$ (reversal in the numerator)
  - $\frac{d^2y}{dx^2}=\frac{(1-2x)(2-6x)-(2y-3x^2)(-2)}{(1-2x)^2}$ (chain rule error)

Model Solution:

$$\frac{d^2y}{dx^2}=\frac{(1-2x)\left(2\frac{dy}{dx}-6x\right)-(2y-3x^2)(-2)}{(1-2x)^2}$$

0	1

The response accurately includes the criteria below.

- Form of quotient rule

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with an incorrect second derivative does not earn **P5**.

Model Solution:

At $(-2,6)$, $\frac{d^2y}{dx^2}=\frac{5(12)-0(-2)}{5^2}=\frac{12}{5}$.

0	1

The response accurately includes the criteria below.

- $\frac{d^2y}{dx^2}$ at $(-2,6)$

Part C Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

3-2-2

- A response with an incorrect second derivative is eligible to earn **P6** for a consistent answer and reason.
- Special cases: The following examples are eligible to earn **P6** for a consistent answer and reason.
  - $\frac{d^2y}{dx^2} = \frac{(2y-3x^2)(-2)-(1-2x)\left(2\frac{dy}{dx}-6x\right)}{(1-2x)^2}$ (reversal in the numerator) In this case, $\frac{d^2y}{dx^2} = -\frac{12}{5} < 0$ at $(-2, 6)$, and therefore a response should conclude that the curve has a relative maximum there.
  - $\frac{d^2y}{dx^2} = \frac{(1-2x)(2-6x)-(2y-3x^2)(-2)}{(1-2x)^2}$ (chain rule error) In this case, $\frac{d^2y}{dx^2} = \frac{14}{5} > 0$ at $(-2, 6)$, and therefore a response should conclude that the curve has a relative minimum there.

Model Solution:

At $(-2, 6)$, $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$, and therefore the curve has a relative minimum there.

0	1

The response accurately includes the criteria below.

- Answer with reason

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response earns **P7** with an equation that involves either $1 \cdot \cos y + x \cdot (\pm \sin y) \cdot \frac{dy}{dx}$ or $1 \cdot \cos y + x \cdot (\pm \sin y)$.

Model Solution:

$$\frac{d}{dx}(y + x\cos y) = \frac{d}{dx}(18)$$

$$\Rightarrow \frac{dy}{dx} + 1 \cdot \cos y + x \cdot (-\sin y) \cdot \frac{dy}{dx} = 0$$

0	1

The response accurately includes the criteria below.

- Product rule

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

3-2-2

Scoring Notes:

- A response earns **P8** with an equation that involves either $\frac{dy}{dx} + 1 \cdot \cos y + x \cdot (\pm \sin y) \cdot \frac{dy}{dx}$ or $\frac{dy}{dx} + 1 \cdot (\pm \sin y) \cdot \frac{dy}{dx}$.

Model Solution:

$$\frac{d}{dx}(y + x\cos y) = \frac{d}{dx}(18)$$
$$\Rightarrow \frac{dy}{dx} + 1 \cdot \cos y + x \cdot (-\sin y) \cdot \frac{dy}{dx} = 0$$

0	1

The response accurately includes the criteria below.

- Implicit differentiation

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned both **P7** and **P8** to be eligible for **P9**.
- **P9** is earned only for a response mathematically equivalent to $\frac{dy}{dx} = \frac{-\cos y}{1 - x\sin y}$.

Model Solution:

$$\Rightarrow \frac{dy}{dx} \cdot (1 - x\sin y) = -\cos y$$
$$\Rightarrow \frac{dy}{dx} = \frac{-\cos y}{1 - x\sin y}$$

0	1

The response accurately includes the criteria below.

- Answer

3-2-2

25. If $\ln(2x+y)=x+1$, then $\frac{dy}{dx}=$

(A) -2

(B) $2x+y-2$

(C) $2x+y$

(D) $4x+2y-2$

(E) $y-\frac{y}{x}$

26. Consider the closed curve in the *xy*-plane given by $x^2+2x+y^4+4y=5$.
Show that dy/dx = −(x+1) / 2(y^3 + 1).

Part A

1 point is earned for implicit differentiation

1 point is earned for verification

$$2x+2+4y^3\frac{dy}{dx}+4\frac{dt}{dx}=0$$

$$(4y^3+4)\frac{dy}{dx}=-2x-2$$

$$\frac{dy}{dx}=\frac{-2(x+1)}{4(y^3+1)}=\frac{-(x+1)}{2(y^3+1)}$$

0	1	2

The student response earns all of the following points:

1 point is earned for implicit differentiation

1 point is earned for verification

$$2x+2+4y^3\frac{dy}{dx}+4\frac{dt}{dx}=0$$

$$(4y^3+4)\frac{dy}{dx}=-2x-2$$

$$\frac{dy}{dx}=\frac{-2(x+1)}{4(y^3+1)}=\frac{-(x+1)}{2(y^3+1)}$$

3-2-2

27. If $x^2 + xy = 10$, then when $x = 2$, $\frac{dy}{dx} =$

(A) $-\frac{7}{2}$ ✓

(B) -2

(C) $\frac{2}{7}$

(D) $\frac{3}{2}$

(E) $\frac{7}{2}$

28. If $x^2 + y^2 = 25$, what is the value of $\frac{d^2y}{dx^2}$ at the point (4,3) ?

(A) $-\frac{25}{27}$ ✓

(B) $-\frac{7}{27}$

(C) $\frac{7}{27}$

(D) $\frac{3}{4}$

(E) $\frac{25}{27}$

3-2-2

29. Consider the curve given by y^2=2+xy.

Show that dy/dx=y/2y−x.

Part A

1 point is earned for the implicit differentiation

1 point is earned if solved for y'

$2yy'=y+xy'$

$(2y-x)y'=y$

$y\prime=\frac{y}{2y-x}$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for the implicit differentiation

1 point is earned if solved for y'

$2yy'=y+xy'$

$(2y-x)y'=y$

$y\prime=\frac{y}{2y-x}$

30. What is the slope of the line tangent to the curve $3y^2$ - $2x^2$ = 6 - $2xy$ at the point (3,2)?

(A) 0

(B) 4/9 ✓

(C) 7/9

(D) 6/7

(E) 5/3

31. In the xy-plane, what is the slope of the line tangent to the graph of $x^2+xy+y^2=7$ at the point $(2,1)$?

(A) $-\frac{4}{3}$

(B) $-\frac{5}{4}$ ✓

(C) −1

(D) $-\frac{4}{5}$

(E) $-\frac{3}{4}$

3-2-2

32. If $x^2y - 3x = y^3 - 3$, then at the point $(-1, 2)$, $\frac{dy}{dx} =$

(A) $-\frac{7}{11}$

(B) $-\frac{7}{13}$

(C) $-\frac{1}{2}$

(D) $-\frac{3}{14}$

(E) 7

33. Given that $3\text{x} - \tan \text{y} = 4$, what is $\frac{dy}{dx}$ in terms of y ?

(A) $\frac{\text{dy}}{\text{dx}}=3\sin^2\text{y}$

(B) $\frac{\text{dy}}{\text{dx}}=3\cos^2\text{y}$

(C) $\frac{\text{dy}}{\text{dx}}=3\text{cosycoty}$

(D) $\frac{dy}{dx} = \frac{3}{1+9y^2}$

34. If $e^x - y = xy^3 + e^2 - 18$, what is the value of dy/dx at the point (2, 2) ?

(A) e^2 - 32

(B) $\frac{e^2-9}{24}$

(C) $\frac{e^2-8}{25}$

(D) $\frac{e^2}{13}$

Answer C

This option is correct. During the implicit differentiation, both the product rule and the chain rule are needed.

3-2-2

35. Let $y = f(x)$ be a twice-differentiable function such that $f(1) = 2$ and $\frac{dy}{dx} = y^3 + 3$. What is the value of $\frac{d^2y}{dx^2}$ at $x = 1$?

(A) 12

(B) 66

(C) 132 ✓

(D) 165

Answer C

This option is correct. The answer is obtained by using the chain rule when doing implicit differentiation.The second derivative is found in terms of y and $\frac{dy}{dx}$, which can then be written completely in terms of y.The condition f (1) = 2 means that y = 2 when x = 1.

$$\frac{d^2y}{dx^2} = 3y^2\frac{dy}{dx} = 3y^2\left(y^3+3\right)$$

$$\frac{d^2y}{dx^2}\bigg|_{(x,y)=(1,2)} = 3.2^2\left(2^3+3\right) = 3\left(4\right)\left(11\right) = 132$$

36. If cos (xy) = y-1, then the value of dy/dx when $x = \pi/2$ and y = 1 is

(A) $\frac{-2}{(2-\pi)}$

(B) $\frac{-2}{(2+\pi)}$ ✓

(C) 0

(D) $\frac{2}{(2-\pi)}$

(E) $\frac{2}{(2+\pi)}$

37. What is the slope of the line tangent to the curve $y + 2 = \frac{x^2}{2} - 2\sin y$ at the point(2,0) ?

(A) -2

(B) 0

(C) $\frac{1}{2}$

(D) $\frac{2}{3}$ ✓

(E) 2

3-2-2

38. If $y^3 + y = x^2$, then $\frac{dy}{dx} =$

(A) 0

(B) $\frac{x}{2}$

(C) $\frac{2x}{3y^2}$

(D) $2x - 3y^2$

(E) $\frac{2x}{1+3y^2}$

39. if $e^{xy} - y^2 = e - 4$, then at $x = 1/2$ and $y = 2$, $dy/dx =$

(A) e/4

(B) e/2

(C) 4e/(8-e)

(D) 4e/(4-e)

(E) (8-4e)/e

40. If $x^2 + xy - 3y = 3$, then at the point (2, 1), $\frac{dy}{dx} =$

(A) 5

(B) 4

(C) $\frac{7}{3}$

(D) 2

41. If $x^2 + xy - 3y = 3$, then at the point (2, 1), $\frac{dy}{dx} =$

(A) 5

(B) 4

(C) $\frac{7}{3}$

(D) 2

42. The slope of the tangent to the curve $y^3x + y^2x^2 = 6$ at (2, 1) is

(A) $-\frac{3}{2}$

(B) -1

(C) $-\frac{5}{14}$

(D) $-\frac{3}{14}$

(E) 0

3-2-2

43. If $x^3 + 3xy + 2y^3 = 17$, then in terms of x and y, $\frac{dy}{dx} =$

(A) $-\frac{x^2+y}{x+2y^2}$

(B) $-\frac{x^2+y}{x+y^2}$

(C) $-\frac{x^2+y}{x+2y}$

(D) $-\frac{x^2+y}{2y^2}$

(E) $\frac{-x^2}{1+2y^2}$

44. Suppose $\ln x - \ln y = y - 4$, where y is a differentiable function of x and $y = 4$ when $x = 4$. What is the value of $\frac{dy}{dx}$ when $x = 4$?

(A) 0

(B) $\frac{1}{5}$

(C) $\frac{1}{3}$

(D) $\frac{1}{2}$

(E) $\frac{17}{5}$

45. If $3x^2 + 2xy + y^2 = 2$, then the value of dy/dx at x = 1 is

(A) -2

(B) 0

(C) 2

(D) 4

(E) not defined

46. If $x + 2xy - y^2 = 2$, then at the point (1,1), $\frac{dy}{dx}$ is

(A) $\frac{3}{2}$

(B) $\frac{1}{2}$

(C) 0

(D) $-\frac{3}{2}$

(E) nonexistent

3-2-2

47. If $\sin(xy) = x$, then $\frac{dy}{dx} =$

(A) $\frac{1}{\cos(xy)}$

(B) $\frac{1}{x\cos(xy)}$

(C) $\frac{1-\cos(xy)}{\cos(xy)}$

(D) $\frac{1-y\cos(xy)}{x\cos(xy)}$

(E) $\frac{y(1-\cos(xy))}{x}$

48. Consider the curve given by $xy^2 - x^3y = 6$.

Show that $\frac{dy}{dx} = \frac{3x^2y-y^2}{2xy-x^3}$.

Part A

1 point is earned for the implicit differentiation

1 point is earned for verifies expression for $\frac{dy}{dx}$

$$y^2 + 2xy\frac{dy}{dx} - 3x^2y - x^3\frac{dy}{dx} = 0$$
$$\frac{dy}{dx}\left(2xy - x^3\right) = 3x^2y - y^2$$
$$\frac{dy}{dx} = \frac{3x^2y-y^2}{2xy-x^3}$$

0	1	2

The student response earns all of the following points:

1 point is earned for the implicit differentiation

1 point is earned for verifies expression for $\frac{dy}{dx}$

$$y^2 + 2xy\frac{dy}{dx} - 3x^2y - x^3\frac{dy}{dx} = 0$$
$$\frac{dy}{dx}\left(2xy - x^3\right) = 3x^2y - y^2$$
$$\frac{dy}{dx} = \frac{3x^2y-y^2}{2xy-x^3}$$

3-2-2

49. If $\sin x = e^y$, $0 < x < \pi$, what is dy/dx in terms of x?

(A) -tan x

(B) -cot x

(C) cot x ✓

(D) tan x

(E) csc x

50. If $(x+2y)\cdot\frac{dy}{dx}=2x-y$, what is the value of $\frac{d^2y}{dx^2}$ at the point $(3,0)$?

(A) $-\frac{10}{3}$ ✓

(B) 0

(C) 2

(D) $\frac{10}{3}$

(E) Undefined

51. If $y=\ln(x^2+y^2)$, then the value of $\frac{dy}{dx}$ at the point (1, 0) is

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) 2 ✓

(E) undefined

3-2-2

52. Consider the curve defined by $2y^3 + 6x^2y - 12x^2 + 6y = 1$.

Show that dy / dx=4x−2xy / x2+y2+1.

Part 6A

1 point is earned for implicit differentiation

1 point is earned if verifies expression for dy/dx

$$6y^2\frac{dy}{dx} + 6x^2\frac{dy}{dx} + 12xy - 24x + 6\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} + \left(6y^2 + 6x^2 + 6\right) = 24x - 12xy$$

$$\frac{dy}{dx} = \frac{24x-12xy}{6x^2+6y^2+6} = \frac{4x-2xy}{x^2+y^2+1}$$

0	1	2 ✓

The student response earns two of the following points:

1 point is earned for implicit differentiation

1 point is earned if verifies expression for dy/dx

$$6y^2\frac{dy}{dx} + 6x^2\frac{dy}{dx} + 12xy - 24x + 6\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} + \left(6y^2 + 6x^2 + 6\right) = 24x - 12xy$$

$$\frac{dy}{dx} = \frac{24x-12xy}{6x^2+6y^2+6} = \frac{4x-2xy}{x^2+y^2+1}$$

53. If $x^2 + xy + y^3 = 0$, then, in terms of x and y, $\frac{dy}{dx}$=

(A) $-\frac{2x+y}{x+3y^2}$ ✓

(B) $-\frac{x+3y^2}{2x+y}$

(C) $\frac{-2x}{1+3y^2}$

(D) $\frac{-2x}{x+3y^2}$

(E) $-\frac{2x+y}{x+3y^2-1}$

3-2-2

54. The slope of the line tangent to the graph of ln*(xy)*=*x* at the point where *x*=1 is

(A) 0 ✓

(B) 1

(C) e

(D) e^2

(E) 1-e

55. If $e^{f(x)} = 1 + x^2$, then $f'(x) =$

(A) $\frac{1}{1+x^2}$

(B) $\frac{2x}{1+x^2}$ ✓

(C) $2x\left(1 + x^2\right)$

(D) $2x\left(e^{1+x^2}\right)$

(E) $2x \ln\left(1 + x^2\right)$

56. If $y = xy + x^2 + 1$, then when *x*=-1, $\frac{dy}{dx}$ is

(A) $\frac{1}{2}$

(B) $-\frac{1}{2}$ ✓

(C) -1

(D) -2

(E) nonexistent

3-2-2

57. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Consider the curve given by the equation $2(x-y)=3+\cos y$. For all points on the curve, $\frac{2}{3}\le\frac{dy}{dx}\le 2$.

(a) Show that $\frac{dy}{dx}=\frac{2}{2-\sin y}$.

(b) For $-\frac{\pi}{2}<y<\frac{\pi}{2}$, there is a point P on the curve through which the line tangent to the curve has slope 1. Find the coordinates of the point P.

(c) Determine the concavity of the curve at points for which $-\frac{\pi}{2}<y<\frac{\pi}{2}$. Give a reason for your answer.

(d) Let $y=f(x)$ be a function, defined implicitly by $2(x-y)=3+\cos y$, that is continuous on the closed interval $[2,\ 2.1]$ and differentiable on the open interval $(2,\ 2.1)$. Use the Mean Value Theorem on the interval $[2,\ 2.1]$ to show that $\frac{1}{15}\le f(2.1)-f(2)\le\frac{1}{5}$.

Part A

1 point is earned for: implicit differentiation

1 point is earned for: verification

$$\frac{d}{dx}(2(x-y))=\frac{d}{dx}(3+\cos y)$$
$$2-2\frac{dy}{dx}=(-\sin y)\frac{dy}{dx}$$
$$2=(2-\sin y)\frac{dy}{dx}$$
$$\frac{dy}{dx}=\frac{2}{2-\sin y}$$

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: implicit differentiation

1 point is earned for: verification

$$\frac{d}{dx}(2(x-y))=\frac{d}{dx}(3+\cos y)$$
$$2-2\frac{dy}{dx}=(-\sin y)\frac{dy}{dx}$$
$$2=(2-\sin y)\frac{dy}{dx}$$
$$\frac{dy}{dx}=\frac{2}{2-\sin y}$$

Part B

1 point is earned for: $\frac{dy}{dx}=1$

3-2-2

1 point is earned for: answer

$$\frac{dy}{dx} = \frac{2}{2 - \sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$$
$$2(x - 0) = 3 + \cos 0 \Rightarrow 2x = 4 \Rightarrow x = 2$$

Point P has coordinates $(2,\ 0)$.

0	1	2

The student response earns all of the following points:

1 point is earned for: $\frac{dy}{dx} = 1$

1 point is earned for: answer

$$\frac{dy}{dx} = \frac{2}{2 - \sin y} = 1 \Rightarrow \sin y = 0 \Rightarrow y = 0$$
$$2(x - 0) = 3 + \cos 0 \Rightarrow 2x = 4 \Rightarrow x = 2$$

Point P has coordinates $(2,\ 0)$.

Part C

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for: answer with reason

$$\frac{d^2y}{dx^2} = \frac{-2}{(2 - \sin y)^2}(-\cos y)\frac{dy}{dx} = \frac{4\cos y}{(2 - \sin y)^3}$$

$\frac{d^2y}{dx^2} > 0$ for all y in the interval $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

Therefore, the curve is concave up for $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

0	1	2	3

The student response earns all of the following points:

2 points are earned for: $\frac{d^2y}{dx^2}$

1 point is earned for: answer with reason

3-2-2

$$\frac{d^2y}{dx^2} = \frac{-2}{(2-\sin y)^2}(-\cos y)\frac{dy}{dx} = \frac{4\cos y}{(2-\sin y)^3}$$

$\frac{d^2y}{dx^2} > 0$ for all y in the interval $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

Therefore, the curve is concave up for $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

Part D

1 point is earned for: applies Mean Value Theorem

1 point is earned for: verification

By the Mean Value Theorem, for some value c in the interval $(2,\ 2.1)$, $f'(c) = \frac{f(2.1)-f(2)}{0.1}$.

For all points on the curve, $\frac{2}{3} \le f'(x) \le 2$.

Thus, $\frac{2}{3} \le \frac{f(2.1)-f(2)}{0.1} \le 2 \Rightarrow \frac{1}{15} \le f(2.1) - f(2) \le \frac{1}{5}$.

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for: applies Mean Value Theorem

1 point is earned for: verification

By the Mean Value Theorem, for some value c in the interval $(2,\ 2.1)$, $f'(c) = \frac{f(2.1)-f(2)}{0.1}$.

For all points on the curve, $\frac{2}{3} \le f'(x) \le 2$.

Thus, $\frac{2}{3} \le \frac{f(2.1)-f(2)}{0.1} \le 2 \Rightarrow \frac{1}{15} \le f(2.1) - f(2) \le \frac{1}{5}$.

58. Given that 3x − tan y = 4, what is $\frac{dy}{dx}$ in terms of y?

(A) $\frac{dy}{dx} = 3\sin^2 y$

(B) $\frac{dy}{dx} = 3\cos^2 y$ ✓

(C) $\frac{dy}{dx} = 3\cos y \cot y$

(D) $\frac{dy}{dx} = \frac{3}{1+9y^2}$

3-2-2

59. The points $(-1,\ -1)$ and $(1,\ -5)$ are on graph of a function $y = f(x)$ that satisfies the differential equation $\frac{dy}{dx} = x^2 + y$. Which of the following must be true?

(A) $(1,\ -5)$ is a local maximum of f.

(B) $(1,\ -5)$ is a point of inflection of the graph of f.

(C) $(-1,\ -1)$ is a local maximum of f. ✓

(D) $(-1,\ -1)$ is a local minimum of f.

(E) $(-1,\ -1)$ is a point of inflection of the graph of f.

3-2-2

60. Consider the closed curve in the xy-plane given by

$$x^2 + 2x + y^4 + 4y = 5.$$

(a) Show that $dy/dx = -(x + 1) / 2(y^3 + 1)$.

(b) Write an equation for the line tangent to the curve at the point (-2, 1).

(c) Find the coordinates of the two points on the curve where the line tangent to the curve is vertical.

(d) Is it possible for this curve to have a horizontal tangent at points where it intersects the x-axis? Explain your reasoning.

Part A

1 point is earned for implicit differentiation

1 point is earned for verification

$$2x + 2 + 4y^3\frac{dy}{dx} + 4\frac{dy}{dx} = 0$$
$$(4y^3 + 4)\frac{dy}{dx} = -2x - 2$$
$$\frac{dy}{dx} = \frac{-2(x+1)}{4(y^3+1)} = \frac{-(x+1)}{2(y^3+1)}$$

0	1	2

The student response earns all of the following points:

1 point is earned for implicit differentiation

1 point is earned for verification

$$2x + 2 + 4y^3\frac{dy}{dx} + 4\frac{dy}{dx} = 0$$
$$(4y^3 + 4)\frac{dy}{dx} = -2x - 2$$
$$\frac{dy}{dx} = \frac{-2(x+1)}{4(y^3+1)} = \frac{-(x+1)}{2(y^3+1)}$$

Part B

1 point is earned for the slope

1 point is earned for the tangent line equation

$$\left.\frac{dy}{dx}\right|_{(-2,1)} = \frac{-(-2+1)}{2(1+1)} = \frac{1}{4}$$

3-2-2

Tangent line: $y=1+\frac{1}{4}(x+2)$

0	1	2

The student response earns all of the following points:

1 point is earned for the slope

1 point is earned for the tangent line equation

$$\frac{dy}{dx}\Big|_{(-2,1)}=\frac{-(-2+1)}{2(1+1)}=\frac{1}{4}$$

Tangent line: $y=1+\frac{1}{4}(x+2)$

Part C

1 point is earned for y = -1

1 point is earned if substitutes y = -1 into the equation of the curve

1 point is earned for the answer

Vertical tangent lines occur at points on the curve where $y^3+1=0$ (or y = -1) and $x\neq-1$.

On the curve, y = -1 implies that $x^2+2x+1-4=5$, so x = -4 or x = 2.

Vertical tangent lines occur at the points (-4, -1) and (2, -1).

0	1	2	3

The student response earns all of the following points:

1 point is earned for y = -1

1 point is earned if substitutes y = -1 into the equation of the curve

1 point is earned for the answer

Vertical tangent lines occur at points on the curve where $y^3+1=0$ (or y = -1) and $x\neq-1$.

On the curve, y = -1 implies that $x^2+2x+1-4=5$, so x = -4 or x = 2.

Vertical tangent lines occur at the points (-4, -1) and (2, -1).

3-2-2

Part D

1 point is earned if works with $x = -1$ or $y = 0$

1 point is earned for answer with reason

Horizontal tangents occur at points on the curve where $x = -1$ and $y \neq -1$.

The curve crosses the x-axis where $y = 0$.

$(-1)^2 + 2(-1) + 0^4 + 4 \cdot 0 \neq 5$

No, the curve cannot have a horizontal tangent where it crosses the x-axis.

0	1	2

The student response earns all of the following points:

1 point is earned if works with $x = -1$ or $y = 0$

1 point is earned for answer with reason

Horizontal tangents occur at points on the curve where $x = -1$ and $y \neq -1$.

The curve crosses the x-axis where $y = 0$.

$(-1)^2 + 2(-1) + 0^4 + 4 \cdot 0 \neq 5$

No, the curve cannot have a horizontal tangent where it crosses the x-axis.

3-2-2

61. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve defined by the equation $xy = x + y^2 + 3$.

(a) Show that $\frac{dy}{dx} = \frac{1-y}{x-2y}$.

(b) Find the slope of the line tangent to the curve at the point $(7, 2)$.

(c) Find an equation for each vertical line tangent to the curve.

(d) Evaluate $\frac{d^2y}{dx^2}$ at the point $(7, 2)$. Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for correct implicit differentiation of $xy = x + y^2 + 3$. Responses may use alternative notation for $\frac{dy}{dx}$, such as y'.

The second point may not be earned without the first point.

It is sufficient to present $(x - 2y)\frac{dy}{dx} = 1 - y$ to earn the second point, provided that there are no subsequent errors.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Implicit differentiation
- ☐ Verification

Solution:

$$\frac{d}{dx}(xy) = \frac{d}{dx}\left(x + y^2 + 3\right) \Rightarrow y + x\frac{dy}{dx} = 1 + 2y\frac{dy}{dx}$$
$$\Rightarrow (x - 2y)\frac{dy}{dx} = 1 - y \Rightarrow \frac{dy}{dx} = \frac{1-y}{x-2y}$$

3-2-2

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{3}$ with no supporting work earns the point.

0	1 ✓

The student response accurately includes the criteria below.

- [ ] Answer

Solution:

Slope $= \left.\frac{dy}{dx}\right|_{(x,\,y)=(7,\,2)} = \frac{1-2}{7-4} = -\frac{1}{3}$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response need not specify that $y-1 \neq 0$ in order to earn any point in this part.

The third point is earned only for both answers $x=-2$ and $x=6$, accompanied by supporting work.

Alternate solution:

Substituting $y=\frac{x}{2}$ in $xy=x+y^2+3$ yields the equation $\frac{x^2}{2}=x+\frac{x^2}{4}+3$.

$\Rightarrow \frac{1}{4}\left(x^2-4x-12\right)=\frac{1}{4}(x+2)(x-6)=0$

$\Rightarrow x=-2$ or $x=6$,
If $x=-2$ then $-2y=-2+y^2+3 \Rightarrow y=-1 \neq 1$.

If $x=6$, then $6y=6+y^2+3 \Rightarrow y=3 \neq 1$.

The vertical tangent lines to the curve have equations $x=-2$ and $x=6$.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- [ ] Sets $x-2y=0$
- [ ] Substitutes $x=2y$ in $xy=x+y^2+3$

3-2-2

☐ Answers

Solution:

For the line tangent to the curve to be vertical, it is necessary that $x - 2y = 0$ (so $x = 2y$) and $y - 1 \neq 0$ ($y \neq 1$).

Substituting $x = 2y$ in $xy = x + y^2 + 3$ yields the equation $2y^2 = 2y + y^2 + 3$.

$\Rightarrow y^2 - 2y - 3 = (y+1)(y-3) = 0$

$\Rightarrow y = -1$ or $y = 3$

If $y = -1$, then $-x = x + 1 + 3 \Rightarrow x = -2$.

If $y = 3$, then $3x = x + 9 + 3 \Rightarrow x = 6$.

The vertical tangent lines to the curve have equations $x = -2$ and $x = 6$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first two points can be earned in either order.

A response that presents at most one error (copy, derivative, missing parenthesis) in applying the quotient rule or product rule earns the first point.

- Any error in a presented expression for $\frac{d^2y}{dx^2}$ is not eligible for the third point.

To earn the second point, a response must present both occurrences of $\frac{dy}{dx}$ (or y') in their expression for $\frac{d^2y}{dx^2}$ (or y'').

A response must have earned both of the first 2 points in order to be eligible for the third point.

The third point is earned with a correct expression for $\left.\frac{d^2y}{dx^2}\right|_{(x,\, y)=(7,\, 2)}$ consistent with the declared value of $\left.\frac{dy}{dx}\right|_{(x,\, y)=(7,\, 2)}$ from part (b).

The value of $\left.\frac{d^2y}{dx^2}\right|_{(x,\, y)=(7,\, 2)}$ does not need to be simplified. Any error in simplification does not earn the third point.

Alternate solution:

$xy = x + y^2 + 3 \Rightarrow y + x\frac{dy}{dx} = 1 + 2y\frac{dy}{dx}$

$\frac{dy}{dx} + x\frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot 1 = 0 + 2y \cdot \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot 2 \cdot \frac{dy}{dx}$

From part (b) $\left.\frac{dy}{dx}\right|_{(x,\, y)=(7,\, 2)} = -\frac{1}{3}$

3-2-2

$$-\tfrac{1}{3}+(7)\frac{d^2y}{dx^2}+\left(-\tfrac{1}{3}\right)=2(2)\cdot\frac{d^2y}{dx^2}+\left(-\tfrac{1}{3}\right)\cdot 2\cdot\left(-\tfrac{1}{3}\right)\Rightarrow\frac{d^2y}{dx^2}=\frac{8}{27}$$

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- ☐ Quotient rule (or product rule)
- ☐ Chain rule
- ☐ Answer

Solution:

$$\frac{dy}{dx}=\frac{1-y}{x-2y}$$

Using the quotient rule, $\dfrac{d^2y}{dx^2}=\dfrac{-\frac{dy}{dx}(x-2y)-(1-y)\left(1-2\frac{dy}{dx}\right)}{(x-2y)^2}$.

From part (b), at $(x,\ y)=(7,\ 2)$, $\left.\dfrac{dy}{dx}\right|_{(x,\ y)=(7,\ 2)}=-\dfrac{1}{3}$.

Thus, $\left.\dfrac{d^2y}{dx^2}\right|_{(x,\ y)=(7,\ 2)}=\dfrac{-\left(-\frac{1}{3}\right)(7-4)-(1-2)\left(1-2\left(-\frac{1}{3}\right)\right)}{(7-4)^2}=\dfrac{1+\frac{5}{3}}{9}=\dfrac{8}{27}$.

3-2-2

62. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which is a real number.

Consider the curve given by the equation $6xy = 2 + y^3$.

(a) Show that $\frac{dy}{dx} = \frac{2y}{y^2-2x}$.

(b) Find the coordinates of a point on the curve at which the line tangent to the curve is horizontal, or explain why no such point exists.

(c) Find the coordinates of a point on the curve at which the line tangent to the curve is vertical, or explain why no such point exists.

(d) A particle is moving along the curve. At the instant when the particle is at the point $\left(\frac{1}{2}, -2\right)$, its horizontal position is increasing at a rate of $\frac{dx}{dt} = \frac{2}{3}$ unit per second. What is the value of $\frac{dy}{dt}$, the rate of change of the particle's vertical position, at that instant?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for the correct implicit differentiation of $6xy = 2 + y^3$. Responses may use alternative notations for $\frac{dy}{dx}$, such as y'.

The second point cannot be earned without the first point.

It is sufficient to present $2y = \frac{dy}{dx}(y^2 - 2x)$ to earn the second point, provided there are no subsequent errors.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Implicit differentiation
- ☐ Verification

Solution:

3-2-2

$\frac{d}{dx}(6xy) = \frac{d}{dx}(2+y^3) \Rightarrow 6y + 6x\frac{dy}{dx} = 3y^2\frac{dy}{dx}$

$\Rightarrow 2y = \frac{dy}{dx}(y^2 - 2x) \Rightarrow \frac{dy}{dx} = \frac{2y}{y^2-2x}$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned with any of $2y = 0$, $y = 0$, $\frac{dy}{dx} = 0$, $\frac{dy}{dx} = 0$, $dy = 0$, $y\prime = 0$, or $\frac{2y}{y^2-2x} = 0$.

A response need not state that at a horizontal tangent, $y^2 - 2x \neq 0$.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Sets $2y = 0$
- [ ] Answer with reason

Solution:

For the line tangent to the curve to be horizontal, it is necessary that $2y = 0$ (so $y = 0$) and that $y^2 - 2x \neq 0$.

Substituting $y = 0$ into $6xy = 2 + y^3$ yields the equation $6x \cdot 0 = 2$, which has no solution.

Therefore, there is no point on the curve at which the line tangent to the curve is horizontal.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned by presenting $y^2 = 2x$ or $y = \sqrt{2x}$.

The second point can be earned for the substitution of $y = \sqrt{2x}$ into $6xy = 2 + y^3$, or for substituting $x = \frac{2+y^3}{6y}$ into $y^2 - 2x = 0$.

A response earns all three points by setting $y^2 - 2x = 0$, declaring the point $(\frac{1}{2}, 1)$, and verifying that this point is on the curve $6xy = 2 + y^3$.

A response that identifies the point $(\frac{1}{2}, 1)$ but does not verify that the point is on the curve, does not earn the second or the third point.

To earn the third point the response must present both coordinates of the point $(\frac{1}{2}, 1)$. The coordinates need not appear as an ordered pair as long as they are labeled.

3-2-2

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Sets $y^2 - 2x = 0$
- [ ] Substitutes $x = \frac{y^2}{2}$ into $6xy = 2 + y^3$
- [ ] Answer

Solution:

For a line tangent to this curve to be vertical, it is necessary that $2y \neq 0$ and that $y^2 - 2x = 0$ (so $x = \frac{y^2}{2}$).

Substituting $x = \frac{y^2}{2}$ into $6xy = 2 + y^3$ yields the equation $3y^2 \cdot y = 2 + y^3 \Rightarrow 2y^3 = 2 \Rightarrow y = 1$.

Substituting $y = 1$ in $6xy = 2 + y^3$ yields $6x = 2 + 1$, or $x = \frac{1}{2}$.

The tangent line to the curve is vertical at the point $(\frac{1}{2}, 1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by presenting one or more of the terms $6y\frac{dx}{dt}$, $6x\frac{dy}{dt}$, or $3y^2\frac{dy}{dt}$.

Units will not affect scoring in this part.

An unsupported response of $-\frac{8}{9}$ earns no points.

Alternate solution:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

$$\left.\frac{dy}{dx}\right|_{(x,\,y)=(\frac{1}{2},\,-2)} = \frac{2(-2)}{(-2)^2 - 2(\frac{1}{2})} = -\frac{4}{3}$$

$$\left.\frac{dy}{dt}\right|_{(x,\,y)=(\frac{1}{2},\,-2)} = \left.\frac{dy}{dx} \cdot \frac{dx}{dt}\right|_{(x,\,y)=(\frac{1}{2},\,-2)} = -\frac{4}{3} \cdot \frac{2}{3} = -\frac{8}{9} \text{ unit per second}$$

- The first point is earned for the statement $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$ or equivalent.
- A numerical expression, such as $-\frac{4}{3} \cdot \frac{2}{3}$ or $\frac{2(-2)}{(-2)^2 - 2(\frac{1}{2})} \cdot \frac{2}{3}$, earns both points.

3-2-2

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Uses implicit differentiation with respect to t
- [ ] Answer

Solution:

$$6y\frac{dx}{dt} + 6x\frac{dy}{dt} = 0 + 3y^2\frac{dy}{dt}$$

At the point $(x, y) = (\frac{1}{2}, -2)$,

$$6(-2)(\tfrac{2}{3}) + 6(\tfrac{1}{2})\frac{dy}{dt} = 3(-2)^2\frac{dy}{dt}$$

$$\Rightarrow -8 + 3\frac{dy}{dt} = 12\frac{dy}{dt}$$

$$\Rightarrow \frac{dy}{dt} = -\tfrac{8}{9} \text{ unit per second}$$

3-2-2

63. Consider the curve given by $y^2 = 2 + xy$.

(a) Show that $\frac{dy}{dx} = \frac{y}{2y-x}$.

(b) Find all points (x, y) on the curve where the line tangent to the curve has slope 1/2.

(c) Show that there are no points (x,y) on the curve where the line tangent to the curve is horizontal.

(d) Let x and y be functions of time t that are related by the equation $y^2 = 2 + xy$. At time $t = 5$, the value of y is 3 and $dy/dt = 6$. Find the value of dx/dt at time $t = 5$.

Part A

1 point is earned for the implicit differentiation

1 point is earned if solved for y'

$2yy' = y + xy'$

$(2y - x)y' = y$

$$y' = \frac{y}{2y - x}$$

0	1	2

The student response earns all of the following points:

1 point is earned for the implicit differentiation

1 point is earned if solved for y'

$2yy' = y + xy'$

$(2y - x)y' = y$

$$y' = \frac{y}{2y - x}$$

Part B

1 point is earned for $\frac{y}{2y-x} = \frac{1}{2}$

1 point is earned for the answer

$\frac{y}{2y-x} = \frac{1}{2}$

3-2-2

$2y = 2y - x$

$x = 0$

$y = \pm\sqrt{2}$

$\left(0, \sqrt{2}\right), \left(0, -\sqrt{2}\right)$

0	1	2

The student response earns all of the following points:

1 point is earned for $\frac{y}{2y-x} = \frac{1}{2}$

1 point is earned for the answer

$\frac{y}{2y-x} = \frac{1}{2}$

$2y = 2y - x$

$x = 0$

$y = \pm\sqrt{2}$

$\left(0, \sqrt{2}\right), \left(0, -\sqrt{2}\right)$

Part C

1 point is earned for $y = 0$

1 point is earned for the explanation

$\frac{y}{2y-x} = 0$

$y = 0$

The curve has no horizontal tangent since

$0^2 \neq 2 + x \cdot 0$ for any x.

3-2-2

0	1	2 ✓

The student response earns all of the following points:

1 point is earned for $y = 0$

1 point is earned for the explanation

$\frac{y}{2y-x} = 0$

$y = 0$

The curve has no horizontal tangent since

$0^2 \neq 2 + x \cdot 0$ for any x.

Part D

1 point is earned if solves for x

1 point is earned for the chain rule

1 point is earned for the answer

When $y = 3$, $3^2 = 2 + 3x$ so $x = 7/3$.

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{y}{2y-x} \cdot \frac{dx}{dt}$$

At $t = 5, 6 = \frac{3}{6-\frac{7}{3}} \cdot \frac{dx}{dt} = \frac{9}{11} \cdot \frac{dx}{dt}$

$$\frac{dx}{dt}|_{t=5} = \frac{22}{3}$$

0	1	2	3 ✓

The student response earns all of the following points:

1 point is earned if solves for x

1 point is earned for the chain rule

1 point is earned for the answer

When $y = 3$, $3^2 = 2 + 3x$ so $x = 7/3$.

3-2-2

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{y}{2y-x} \cdot \frac{dx}{dt}$$

At $t = 5, 6 = \frac{3}{6-\frac{7}{3}} \cdot \frac{dx}{dt} = \frac{9}{11} \cdot \frac{dx}{dt}$

$$\frac{dx}{dt}\Big|_{t=5} = \frac{22}{3}$$

3-2-2

64. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve given by $x^2 - xy + 2y^2 = 7$.

(a) Show that $\frac{dy}{dx} = \frac{y-2x}{4y-x}$.

(b) Determine the y-coordinate of each point on the curve at which the line tangent to the curve at that point is vertical. Justify your answer.

(c) Find $\frac{d^2y}{dx^2}$ in terms of x, y, and $\frac{dy}{dx}$. The line tangent to the curve at the point $(1, 2)$ is horizontal. Determine whether the curve is concave up or concave down at the point $(1, 2)$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Responses may use alternative notations for $\dfrac{dy}{dx}$ such as y'.

The first point is earned if there is at most one sign error for the derivative of the term $-xy$ and the remainder of the equation is correct.

The first point is earned with a completely correct derivative of $x^2 - xy + 2y^2$.

The second point cannot be earned if there are any errors.

It is sufficient to present $(4y - x)\dfrac{dy}{dx} = y - 2x$ to earn the second point, provided that there are no subsequent errors. It is not necessary to simplify to $\dfrac{dy}{dx} = \dfrac{y-2x}{4y-x}$.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Implicit differentiation
- [ ] Verification

Solution:

3-2-2

$$2x - \left(y + x\frac{dy}{dx}\right) + 4y\frac{dy}{dx} = 0$$

$$\Rightarrow\ 4y\frac{dy}{dx} - x\frac{dy}{dx} = y - 2x \Rightarrow (4y - x)\frac{dy}{dx} = y - 2x$$
$$\Rightarrow\ \frac{dy}{dx} = \frac{y-2x}{4y-x}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The second point is earned by substituting $x = 4y$ or $y = \frac{1}{4}x$ into the equation of the curve.

$y - 2x \neq 0$ is not required to earn the first point, but it is required to earn the third point.

A response that imports an incorrect expression for $\frac{dy}{dx}$ from part (a) can earn the first and second points only if the expression for $\frac{dy}{dx}$ is a quotient of two linear polynomials in x and y with both variables appearing in the numerator and denominator. Such a response cannot earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] Sets $4y - x = 0$
- [ ] Substitution of $x = 4y$
- [ ] Answer with justification

Solution:

The line tangent to the curve is vertical when $4y - x = 0$ and $y - 2x \neq 0$.

$$4y - x = 0\ \Rightarrow\ x = 4y$$

$$(4y)^2 - (4y)y + 2y^2 = 7$$

$$\Rightarrow\ 16y^2 - 4y^2 + 2y^2 = 7$$
$$\Rightarrow\ 14y^2 = 7\ \Rightarrow\ y^2 = \frac{7}{14}$$
$$\Rightarrow\ y = -\sqrt{\frac{1}{2}}\ \text{or}\ y = \sqrt{\frac{1}{2}}$$

Because $x = 4y$ and $y \neq 0$, $y - 2x \neq 0$.

Therefore, the lines tangent to the curve at $y = \pm\sqrt{\frac{1}{2}}$ are vertical.

Part C

3-2-2

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a correct application of the quotient rule (or product rule), with or without the chain rule.

The second point is earned if the derivatives of all of the y terms are correct (i.e., $\frac{dy}{dx}$ and $4\frac{dy}{dx}$) AND the presented $\frac{d^2y}{dx^2}$ satisfies exactly one of the following:

· $\frac{d^2y}{dx^2}$ is completely correct.

· The quotient rule was used and

o there are sign errors in front of either or both of the two terms in the numerator of $\frac{d^2y}{dx^2}$, and there are no other errors.

o the numerator of $\frac{d^2y}{dx^2}$ is correct but the denominator is not squared, and there are no other errors.

o the numerator of $\frac{d^2y}{dx^2}$ has missing parentheses around the factors in one or both terms, and there are no other errors.

· The product rule was used and there is a sign error between the two terms, and there are no other errors.

· The derivative of one of either $-2x$ or $-x$ is incorrect, and there are no other errors.

To earn the third point, a response must have earned at least 1 of the first 2 points, presented a value of $\frac{d^2y}{dx^2}$ at $(1,\ 2)$, and have a conclusion consistent with this presented value.

To earn the fourth point, a response must have earned the first 3 points and must present a $\frac{d^2y}{dx^2}$ equivalent to $-\frac{2}{7}$ with no computational errors.

A response that imports an incorrect expression for $\frac{dy}{dx}$ from part (a) can earn the first and second points only if the expression for $\frac{dy}{dx}$ is a quotient of two linear polynomials in x and y with both variables appearing in the numerator and denominator. Such a response cannot earn the third or fourth points.

0	1	2	3	4 ✓

The student response accurately includes **all four** of the criteria below.

- ☐ Quotient rule (or product rule)
- ☐ Chain rule
- ☐ Answer
- ☐ Reason

Solution:

3-2-2

$$\frac{d^2y}{dx^2} = \frac{(4y-x)\left(\frac{dy}{dx}-2\right)-(y-2x)\left(4\frac{dy}{dx}-1\right)}{(4y-x)^2}$$

Because the line tangent to the curve at $(1,\ 2)$ is horizontal, $\left.\frac{dy}{dx}\right|_{(x,\ y)=(1,\ 2)} = 0$.

Therefore,

$$\left.\frac{d^2y}{dx^2}\right|_{(x,y)=(1,2)} = \frac{(8-1)(0-2)-(2-2)(0-1)}{(8-1)^2} = -\frac{14}{49} = -\frac{2}{7}.$$

Because $\left.\frac{d^2y}{dx^2}\right|_{(x,\ y)=(1,\ 2)} < 0$, the curve is concave down at the point $(1,\ 2)$.

3-2-2

65. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the function $y = f(x)$ whose curve is given by the equation $2y^2 - 6 = y\sin x$ for $y > 0$.

(a) Show that $\frac{dy}{dx} = \frac{y\cos x}{4y - \sin x}$.

(b) Write an equation for the line tangent to the curve at the point $(0, \sqrt{3})$.

(c) For $0 \leq x \leq \pi$ and $y > 0$, find the coordinates of the point where the line tangent to the curve is horizontal.

(d) Determine whether f has a relative minimum, a relative maximum, or neither at the point found in part (c). Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned only for correctly implicitly differentiating $2y^2 - 6 = y\sin x$. Responses may use alternative notations for $\frac{dy}{dx}$, such as y'.

The second point may not be earned without the first point.

It is sufficient to present $\frac{dy}{dx}(4y - \sin x) = y\cos x$ to earn the second point, provided that there are no subsequent errors.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Implicit differentiation
- ☐ Verification

Solution:

$$\frac{d}{dx}(2y^2 - 6) = \frac{d}{dx}(y\sin x) \Rightarrow 4y\frac{dy}{dx} = \frac{dy}{dx}\sin x + y\cos x$$

3-2-2

$\Rightarrow\ 4y\frac{dy}{dx} - \frac{dy}{dx}\sin\ x = y\ \cos\ x\ \Rightarrow\ \frac{dy}{dx}(4y - \sin\ x) = y\ \cos\ x$

$\Rightarrow\ \frac{dy}{dx} = \frac{y\ \cos\ x}{4y - \sin\ x}$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Any correct tangent line equation will earn the point. No supporting work is required. Simplification of the slope value is not required.

0	1 ✓

The student response accurately includes the criteria below.

☐ Answer

Solution:

At the point $\left(0,\ \sqrt{3}\right)$, $\frac{dy}{dx} = \frac{\sqrt{3}\cos\ 0}{4\sqrt{3} - \sin\ 0} = \frac{1}{4}$.

An equation for the tangent line is $y = \sqrt{3} + \frac{1}{4}x$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by any of $\frac{dy}{dx} = 0$, $\frac{y\ \cos\ x}{4y - \sin\ x} = 0$, or $\cos\ x = 0$.

If additional "correct" x-values are considered outside of the given domain, the response must commit to only $x = \frac{\pi}{2}$ to earn the second point. Any presented y-values, correct or incorrect, are not considered for the second point.

Entering with $x = \frac{\pi}{2}$ does not earn the first point, earns the second point, and is eligible for the third point. The third point is earned for finding $y = 2$. The coordinates do not have to be presented as an ordered pair.

The third point is not earned with additional points present unless the response commits to the correct point.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

☐ Sets $\frac{dy}{dx} = 0$

3-2-2

- ☐ $x = \frac{\pi}{2}$
- ☐ $y = 2$

Solution:

$\frac{dy}{dx} = \frac{y \cos x}{4y - \sin x} = 0 \;\Rightarrow y \cos x = 0$ and $4y - \sin x \neq 0$

$y \cos x = 0$ and $y > 0 \Rightarrow \; x = \frac{\pi}{2}$

When $x = \frac{\pi}{2}$, $y \sin x = 2y^2 - 6 \;\Rightarrow y \sin \frac{\pi}{2} = 2y^2 - 6$
$\Rightarrow \; y = 2y^2 - 6 \Rightarrow \; 2y^2 - y - 6 = 0$
$\Rightarrow \; (2y + 3)(y - 2) = 0 \;\Rightarrow\; y = 2$

When $x = \frac{\pi}{2}$ and $y = 2$, $4y - \sin x = 8 - 1 \neq 0$. Therefore, the line tangent to the curve is horizontal at the point $\left(\frac{\pi}{2},\ 2\right)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an attempt to use the quotient rule (or product rule) to find $\frac{d^2y}{dx^2}$.

The second point is earned for correctly finding $\frac{d^2y}{dx^2}$ and evaluating to find that $\frac{d^2y}{dx^2} < 0$ at $\left(\frac{\pi}{2},\ 2\right)$. The explicit value of $-\frac{2}{7}$ or the equivalent does not need to be reported, but any reported values must be correct in order to earn this point.

The third point can be earned without the second point by reaching a consistent conclusion based on the reported sign of a nonzero value of $\frac{d^2y}{dx^2}$ obtained utilizing $\frac{dy}{dx} = 0$.

Imports: A response is eligible to earn all 3 points in part (d) with a point of the form $\left(\frac{\pi}{2},\ k\right)$ with $k > 0$, imported from part (c).

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $\frac{d^2y}{dx^2}$
- ☐ $\frac{d^2y}{dx^2}$ at $\left(\frac{\pi}{2},\ 2\right)$
- ☐ Answer with justification

Solution:

3-2-2

$$\frac{d^2y}{dx^2} = \frac{(4y - \sin x)\left(\frac{dy}{dx}\cos x - y\sin x\right) - (y\cos x)\left(4\frac{dy}{dx} - \cos x\right)}{(4y - \sin x)^2}$$

When $x = \frac{\pi}{2}$ and $y = 2$,

$$\frac{d^2y}{dx^2} = \frac{\left(4\cdot 2 - \sin\frac{\pi}{2}\right)\left(0\cdot\cos\frac{\pi}{2} - 2\cdot\sin\frac{\pi}{2}\right) - \left(2\cos\frac{\pi}{2}\right)\left(4\cdot 0 - \cos\frac{\pi}{2}\right)}{\left(4\cdot 2 - \sin\frac{\pi}{2}\right)^2}$$

$$= \frac{(7)(-2) - (0)(0)}{(7)^2} = \frac{-2}{7} < 0.$$

f has a relative maximum at the point $\left(\frac{\pi}{2},\ 2\right)$ because $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} < 0$.

3-2-2

66. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve defined by the equation $y^2 = 3xy - 3x$.

(a) Show that $\frac{dy}{dx} = \frac{3y-3}{2y-3x}$.

(b) Find $\frac{d^2y}{dx^2}$ in terms of x, y, and $\frac{dy}{dx}$.

(c) Is there a point on the curve at which the line tangent to the curve is horizontal? Explain your reasoning.

(d) A particle travels along the curve. The line tangent to the path of the particle is vertical at the point P. What are all possible coordinates (x, y) for point P ?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Responses may use other notation for $\frac{dy}{dx}$, including y' and dy.

The second point cannot be earned without the first.

It is sufficient to present $\frac{dy}{dx}(2y - 3x) = 3y - 3$ to earn the second point, provided that there are no subsequent errors.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ Implicit differentiation
- ☐ Verification

Solution:

3-2-2

$$\frac{d}{dx}(y^2) = \frac{d}{dx}(3xy - 3x) \Rightarrow 2y\frac{dy}{dx} = 3x\frac{dy}{dx} + 3y - 3$$
$$\Rightarrow 2y\frac{dy}{dx} - 3x\frac{dy}{dx} = 3y - 3 \Rightarrow \frac{dy}{dx}(2y - 3x) = 3y - 3$$
$$\Rightarrow \frac{dy}{dx} = \frac{3y-3}{2y-3x}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a correct application of the quotient rule, with or without the chain rule: $\frac{(2y-3x) \cdot A - (3y-3) \cdot B}{(2y-3x)^2}$, where $A = \left(3\frac{dy}{dx}\right)$ or $A = (3)$, and $B = \left(2\frac{dy}{dx} - 3\right)$ or $B = (2 - 3)$ (unsimplified).

The second point is earned when the correct second derivative (in terms of x, y, and $\frac{dy}{dx}$) is presented. Subsequent attempts at algebraic simplification do not change the points earned in this part.

Alternate solution to part (b):

· From part (a), $2y\frac{dy}{dx} = 3x\frac{dy}{dx} + 3y - 3$.

· Implicit differentiation: $2yy'' + 2(y')^2 = 3xy'' + 3y' + 3y'$, which earns the first point for correct application of the product rule, with or without chain rule. That is, $2yy'' + 2(y')^2 = 3xy'' + 3y' + 3$ would earn the first point, even with the chain rule error in the final term.

· Solving for y'' : $(2y - 3x)y'' = 6y' - 2(y')^2$, which implies $y'' = \frac{6y' - 2(y')^2}{(2y-3x)}$. This is a correct expression for the second derivative in terms of x, y, and $\frac{dy}{dx}$, and earns the second point. Subsequent attempts at algebraic simplification do not change the points earned in this part.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] Quotient rule
- [ ] $\frac{d^2y}{dx^2}$

Solution:

$$\frac{d^2y}{dx^2} = \frac{(2y-3x)\left(3\frac{dy}{dx}\right) - (3y-3)\left(2\frac{dy}{dx} - 3\right)}{(2y-3x)^2}$$

Part C

3-2-2

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

If a solution begins with $y = 1$, the student does not earn the first point but is eligible for the second point.

0	1	2 ✓

The student response accurately includes **both** of the criteria below.

- ☐ Considers $\frac{dy}{dx} = 0$
- ☐ Answer with explanation

Solution:

$$\frac{dy}{dx} = \frac{3y-3}{2y-3x} = 0 \;\Rightarrow\; y = 1$$

However, when $y = 1$, $1^2 = 3x \cdot 1 - 3x \;\Rightarrow\; 1 = 0$.

Therefore, there is no point on the curve where $y = 1$; thus, there is no horizontal tangent line.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The second and third points cannot be earned without the first point.

A response that earns the first point and finds both $x = 0$ and $x = \frac{4}{3}$, but fails to find either or both y-coordinates, will earn two points in this part.

If more than two ordered pairs are given, the third point is not earned.

0	1	2	3 ✓

The student response accurately includes **all three** of the criteria below.

- ☐ Considers $2y - 3x = 0$
- ☐ Both x-coordinates
  – OR –
  Both y-coordinates
  – OR –
  One coordinate pair
- ☐ Second coordinate pair

3-2-2

Solution:

$2y - 3x = 0 \;\Rightarrow\; y = \frac{3}{2}x$

When $y = \frac{3}{2}x$,

$\left(\frac{3}{2}x\right)^2 = 3x \cdot \frac{3}{2}x - 3x \;\Rightarrow\; \frac{9}{4}x^2 - 3x = 0$
$\Rightarrow\; 3x\left(\frac{3}{4}x - 1\right) = 0 \;\Rightarrow\; x = 0 \text{ or } x = \frac{4}{3}$

When $x = 0$, $y^2 = 0 - 0 \;\Rightarrow\; y = 0$.

When $x = \frac{4}{3}$, $y^2 = 4y - 4 \;\Rightarrow\; y^2 - 4y + 4 = 0 \;\Rightarrow\; y = 2$.

The coordinates $(x,\ y)$ for point P could be $(0,\ 0)$ or $\left(\frac{4}{3},\ 2\right)$.

3-2-2

67. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve defined by $2x^2 + 3y^2 - 4xy = 36$.

(a) Show that $\frac{dy}{dx} = \frac{2y-2x}{3y-2x}$.

(b) Find the slope of the line tangent to the curve at each point on the curve where $x = 6$.

(c) Find the positive value of x at which the curve has a vertical tangent line. Show the work that leads to your answer.

(d) Let x and y be functions of time t that are related by the equation $2x^2 + 3y^2 - 4xy = 36$. At time $t = 1$, the value of x is 2, the value of y is -2, and the value of $\frac{dy}{dt}$ is 4. Find the value of $\frac{dx}{dt}$ at time $t = 1$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned by showing implicit differentiation. Any errors in the response due to distributing, such as negative signs, are part of the second point. Responses that do not correctly differentiate the 36 do not earn the first point.

0	1	2

The student response accurately includes **both** of the criteria below.

- [ ] implicit differentiation
- [ ] verification

Solution:

$$\frac{d}{dx}\left(2x^2 + 3y^2 - 4xy\right) = \frac{d}{dx}(36) \;\Rightarrow\; 4x + 6y\frac{dy}{dx} - 4y - 4x\frac{dy}{dx} = 0$$

$$\Rightarrow\; (6y - 4x)\frac{dy}{dx} = 4y - 4x \;\Rightarrow\; \frac{dy}{dx} = \frac{4y-4x}{6y-4x} = \frac{2y-2x}{3y-2x}$$

Part B

3-2-2

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Simplification is not required to earn either point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ slope at $(6,\ 2)$
- ☐ slope at $(6,\ 6)$

Solution:

$$x = 6 \ \Rightarrow\ 2 \cdot 6^2 + 3y^2 - 4 \cdot 6 \cdot y = 36$$
$$\Rightarrow\ y^2 - 8y + 12 = 0 \ \Rightarrow\ y = 2 \text{ or } y = 6$$

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=(6,\ 2)} = \frac{4-12}{6-12} = \frac{4}{3}$$

The slope of the line tangent to the curve at $(6,\ 2)$ is $\frac{4}{3}$.

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=(6,\ 6)} = \frac{12-12}{18-12} = 0$$

The slope of the line tangent to the curve at $(6,\ 6)$ is 0.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an equation equivalent to $3y - 2x = 0$.

Simplification is not required to earn the second point. Responses that include negative answers, such as $-3\sqrt{6}$ or $\pm\sqrt{54}$, do not earn the second point.

0	1	2

The student response accurately includes **both** of the criteria below.

- ☐ sets $3y - 2x = 0$
- ☐ answer

Solution:

3-2-2

The curve has vertical tangent lines where $3y - 2x = 0$ and $2y - 2x \neq 0$.

$$3y - 2x = 0 \;\Rightarrow\; y = \tfrac{2}{3}x \;\Rightarrow\; 2x^2 + 3\left(\tfrac{2}{3}x\right)^2 - 4x \cdot \tfrac{2}{3}x = 36$$
$$\Rightarrow\; \left(2 + \tfrac{4}{3} - \tfrac{8}{3}\right)x^2 = \tfrac{2}{3}x^2 = 36 \;\Rightarrow\; x^2 = 54$$

Because x is positive, $x = 3\sqrt{6}$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned with work that shows evidence of using the product rule or the chain rule with respect to t.

In order to earn the second point, the first point must have been earned, and the response includes a complete and correct solution for the derivative with respect to t.

Simplification is not required to earn the third point.

0	1	2	3

The student response accurately includes **all three** of the criteria below.

- [ ] chain rule or product rule
- [ ] derivative with respect to t
- [ ] answer

Solution:

$$\frac{d}{dt}\left(2x^2 + 3y^2 - 4xy\right) = \frac{d}{dt}(36)$$
$$\Rightarrow\; 4x\frac{dx}{dt} + 6y\frac{dy}{dt} - 4\left(x\frac{dy}{dt} + y\frac{dx}{dt}\right) = 0$$
$$\Rightarrow\; (4x - 4y)\frac{dx}{dt} + (6y - 4x)\frac{dy}{dt} = 0$$

$$(4 \cdot 2 - 4 \cdot (-2))\frac{dx}{dt}\Big|_{t=1} + (6 \cdot (-2) - 4 \cdot 2) \cdot 4 = 0 \;\Rightarrow\; \frac{dx}{dt}\Big|_{t=1} = 5$$

OR

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{2y - 2x}{3y - 2x} \cdot \frac{dx}{dt}$$

$$4 = \frac{2 \cdot (-2) - 2 \cdot 2}{3 \cdot (-2) - 2 \cdot 2} \cdot \frac{dx}{dt}\Big|_{t=1} = \frac{4}{5} \cdot \frac{dx}{dt}\Big|_{t=1} \;\Rightarrow\; \frac{dx}{dt}\Big|_{t=1} = 5$$

3-2-2

68. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve given by the equation $2xy + y^2 = 8$ for $y > 0$.

(a) Show that $\frac{dy}{dx} = \frac{-y}{x+y}$.

(b) Write an equation for the line tangent to the curve at the point $(1, 2)$.

(c) Evaluate $\frac{d^2y}{dx^2}$ at the point $(1, 2)$.

(d) The points $(1, 2)$ and $\left(\frac{7}{2}, 1\right)$ are on the curve. Find the value of $\left(y^{-1}\right)'(1)$.

Part A

The first point is earned for $2x\frac{dy}{dx} + 2y + 2y\frac{dy}{dx} = 0$ or equivalent

The second point cannot be earned without the first point. The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] implicit differentiation
- [ ] verification

Solution:

$$\frac{d}{dx}\left(2xy + y^2\right) = \frac{d}{dx}(8)$$
$$\Rightarrow\ 2x\frac{dy}{dx} + 2y + 2y\frac{dy}{dx} = 0$$
$$\Rightarrow\ x\frac{dy}{dx} + y\frac{dy}{dx} = -y$$
$$\Rightarrow\ \frac{dy}{dx} = \frac{-y}{x+y}$$

3-2-2

Part B

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below

- ☐ slope
- ☐ equation for tangent line

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=(1,\ 2)} = \frac{-2}{1+2} = -\frac{2}{3}$$

An equation for the tangent line is $y = 2 - \frac{2}{3}(x-1)$.

Part C

The first point is earned with

$$\frac{(x+y)\cdot\left(-1\frac{dy}{dx}\right)-(-y)\left(1+1\frac{dy}{dx}\right)}{(x+y)^2}$$ or equivalent.

The second point is earned with an algebraic or numerical substitution for $\frac{dy}{dx}$.

The response does not require an explicit expression for $\frac{d^2y}{dx^2}$.

The third point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3 ✓

The student responses accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

3-2-2

Solution:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{-y}{x+y}\right)$$

$$= \frac{(x+y)\cdot\left(-1\frac{dy}{dx}\right) - (-y)\left(1+1\frac{dy}{dx}\right)}{(x+y)^2}$$

$$= \frac{(x+y)\cdot\left(\frac{y}{x+y}\right) + y\left(1+\left(\frac{-y}{x+y}\right)\right)}{(x+y)^2} = \frac{y^2+2xy}{(x+y)^3}$$

$$\left.\frac{d^2y}{dx^2}\right|_{(x,\ y)=(1,\ 2)} = \frac{2^2+2(1)(2)}{(1+2)^3} = \frac{8}{27}$$

Part D

The second point may be earned if response contains an incorrect value based on one computational error. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response does not accurately include any of the criteria below.

- ☐ $\frac{dy}{dx}$ at $\left(\frac{7}{2},\ 1\right)$
- ☐ answer

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=\left(\frac{7}{2},\ 1\right)} = \frac{-1}{\left(\frac{7}{2}+1\right)} = -\frac{2}{9}$$

Because $\left(\frac{7}{2},\ 1\right)$ is on the curve, $\left(1,\ \frac{7}{2}\right)$ is on the inverse.

$$\left(y^{-1}\right)'(1) = \frac{1}{y'\left(y^{-1}(1)\right)} = \frac{1}{y'\left(\frac{7}{2}\right)} = -\frac{9}{2}$$

$$\left(y^{-1}\right)'(1) = \frac{1}{y'\left(y^{-1}(1)\right)} = \frac{1}{y'\left(\frac{7}{2}\right)} = -\frac{9}{2}$$

3-2-2

69. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve given by the equation $(2y+1)^3 - 24x = -3$.

(a) Show that $\frac{dy}{dx} = \frac{4}{(2y+1)^2}$.

(b) Write an equation for the line tangent to the curve at the point $(-1, -2)$.

(c) Evaluate $\frac{d^2y}{dx^2}$ at the point $(-1, -2)$.

(d) The point $\left(\frac{1}{6}, 0\right)$ is on the curve. Find the value of $\left(y^{-1}\right)'(0)$.

Part A

The first point is earned for $3(2y+1)^2 \cdot 2\frac{dy}{dx} - 24 = 0$ or equivalent.

The second point cannot be earned without the first point. The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] implicit differentiation
- [ ] verification

Solution:

$$\frac{d}{dx}\left((2y+1)^3 - 24x\right) = \frac{d}{dx}(-3)$$

$$\Rightarrow\ 3(2y+1)^2 \cdot 2\frac{dy}{dx} - 24 = 0$$

$$\Rightarrow\ \frac{dy}{dx} = \frac{24}{6(2y+1)^2} = \frac{4}{(2y+1)^2}$$

3-2-2

Part B

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- [ ] slope
- [ ] equation for tangent line

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\,y)=(-1,\,-2)} = \frac{4}{(2(-2)+1)^2} = \frac{4}{9}$$

An equation for the tangent line is $y = -2 + \frac{4}{9}(x+1)$.

Part C

The first point is earned with $4\left(-2\right)(2y+1)^{-3} \cdot 2\frac{dy}{dx}$ or equivalent.

The second point is earned with an algebraic or numerical substitution for $\frac{dy}{dx}$. The response does not require an explicit expression for $\frac{d^2y}{dx^2}$. The third point does not require a simplified answer.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2	3 ✓

The student responses accurately includes all three of the criteria below.

- [ ] implicit differentiation (2 points for full credit)
- [ ] answer

Solution:

3-2-2

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(4(2y+1)^{-2}\right)$$

$$= 4\,(-2)(2y+1)^{-3} \cdot 2\frac{dy}{dx}$$

$$= \frac{-16}{(2y+1)^3} \cdot \frac{4}{(2y+1)^2} = \frac{-64}{(2y+1)^5}$$

$$\left.\frac{d^2y}{dx^2}\right|_{(x,\ y)=(-1,\ -2)} = \frac{-64}{(2(-2)+1)^5} = \frac{-64}{-243} = \frac{64}{243}$$

Part D

The second point may be earned if response contains an incorrect value based on one computational error. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] $\frac{dy}{dx}$ at $\left(\frac{1}{6},\ 0\right)$
- [ ] answer

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=\left(\frac{1}{6},\ 0\right)} = \frac{4}{(2\cdot 0+1)^2} = 4$$

Because $\left(\frac{1}{6},\ 0\right)$ is on the curve, $\left(0,\ \frac{1}{6}\right)$ is on the inverse.

$$\left(y^{-1}\right)'(0) = \frac{1}{y'\left(y^{-1}(0)\right)} = \frac{1}{y'\left(\frac{1}{6}\right)} = \frac{1}{4}$$

3-2-2

70. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve defined by $y^2 - x^2y = 6$ for $y > 0$.

(a) Show that $\frac{dy}{dx} = \frac{2xy}{2y-x^2}$.

(b) Write an equation for the line tangent to the curve at the point $(1, 3)$.

(c) Show that there is a point P with x-coordinate 0 at which the line tangent to the curve at P is horizontal. Find the y-coordinate of P.

(d) Find the value of $\frac{d^2y}{dx^2}$ at the point P found in part (c).

Part A

The first point is earned for $2y\dfrac{dy}{dx} - 2xy - x^2\dfrac{dy}{dx} = 0$ or equivalent.

The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ implicit differentiation
- ☐ verification

Solution:

3-2-2

$$\frac{d}{dx}\left(y^2 - x^2y\right) = \frac{d}{dx}(6)$$
$$\Rightarrow\ 2y\frac{dy}{dx} - 2xy - x^2\frac{dy}{dx} = 0$$
$$\Rightarrow\ \left(2y - x^2\right)\frac{dy}{dx} = 2xy$$
$$\Rightarrow\ \frac{dy}{dx} = \frac{2xy}{2y - x^2}$$

Part B

The second point may be earned if response contains an incorrect value for slope based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ slope
- ☐ equation for tangent line

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\,y)=(1,\,3)} = \frac{2\cdot1\cdot3}{2\cdot3 - 1^2} = \frac{6}{5}$$

An equation for the tangent line is $y = 3 + \frac{6}{5}(x - 1)$.

Part C

The second point may be earned if response contains an incorrect value for the y- coordinate based on one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2 ✓

The student response accurately includes both of the criteria below.

- ☐ y- coordinate
- ☐ verification

Solution:

3-2-2

At the point P with $x = 0$, it follows that $y^2 - 0 \cdot y = 6 \Rightarrow y = \sqrt{6}$.

$$\left.\frac{dy}{dx}\right|_{(x,\, y)=\left(0,\, \sqrt{6}\right)} = \frac{2 \cdot 0 \cdot \sqrt{6}}{2 \cdot \sqrt{6} - 0^2} = 0$$

Therefore, the line tangent to the curve at P is horizontal.

Part D

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the quotient rule and chain rule with a maximum of one computational error.

The third point requires substitution of values with a consistent use of the y- coordinate from part (c). A simplified answer is not required although the expression should simplify to 1 regardless of the y- coordinate value.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3

The student response accurately includes all three of the criteria below.

- [ ] implicit differentiation (2 points for full credit)
- [ ] answer

Solution:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{2xy}{2y - x^2}\right) = \frac{\left(2y + 2x\frac{dy}{dx}\right)\left(2y - x^2\right) - 2xy\left(2\frac{dy}{dx} - 2x\right)}{\left(2y - x^2\right)^2}$$

$$\left.\frac{d^2y}{dx^2}\right|_{(x,\, y)=\left(0,\, \sqrt{6}\right)} = \frac{\left(2\sqrt{6} + 0\right)\left(2\sqrt{6} - 0\right) - 0}{\left(2\sqrt{6} - 0\right)^2} = \frac{\left(2\sqrt{6}\right)^2}{\left(2\sqrt{6}\right)^2} = 1$$

3-2-2

71. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Consider the curve defined by $xy^2 - 2x^3 = 2$ for $y \geq 0$.

(a) Show that $\frac{dy}{dx} = \frac{6x^2 - y^2}{2xy}$.

(b) Write an equation for the line tangent to the curve at the point $(1, 2)$.

(c) Find the x-coordinate of the point P at which the line tangent to the curve at P is horizontal.

(d) Find the value of $\frac{d^2y}{dx^2}$ at the point $(1, 2)$.

Part A

The first point is earned for $x \cdot 2y\frac{dy}{dx} + 1 \cdot y^2 - 6x^2 = 0$ or equivalent. The second point is earned for a response that arrives at the given expression rather than an algebraically equivalent expression. Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] implicit differentiation
- [ ] verification

Solution:

$$\frac{d}{dx}\left(xy^2 - 2x^3\right) = \frac{d}{dx}(2)$$

$$\Rightarrow \; x \cdot 2y\frac{dy}{dx} + 1 \cdot y^2 - 6x^2 = 0$$

$$\Rightarrow \; 2xy\frac{dy}{dx} = 6x^2 - y^2$$

$$\Rightarrow \; \frac{dy}{dx} = \frac{6x^2 - y^2}{2xy}$$

Part B

3-2-2

The second point may be earned if response contains an incorrect value for slope based on one computational error.Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2

The student response accurately includes both of the criteria below.

- [ ] slope
- [ ] equation for tangent line

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=(1,\ 2)} = \frac{6(1)^2-(2)^2}{2(1)(2)} = \frac{2}{4} = \frac{1}{2}$$

An equation for the tangent line is $y = 2 + \frac{1}{2}(x-1)$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

0	1	2

The student response does not accurately include any of the criteria below.

- [ ] $6x^2 - y^2 = 0$
- [ ] x -coordinate

Solution:

If the line tangent to the curve at P is horizontal, $\frac{dy}{dx} = 0.$

Therefore,$6x^2 - y^2 = 0,\ x \neq 0,\ y \neq 0.$

Because $xy^2 - 2x^3 = 2,$, $6x^2 - \left(\frac{2x^3+2}{x}\right) = 0.$

$$6x^2 - \left(\frac{2x^3+2}{x}\right) = 0 \Rightarrow 6x^3 - 2x^3 = 2 \Rightarrow 4x^3 = 2 \Rightarrow x = \frac{1}{\sqrt[3]{2}}$$

Part D

3-2-2

At most 1 out of 2 implicit differentiation points is earned for correct implicit differentiation with a correct application of the quotient rule and chain rule with a maximum of one computational error.

The third point requires substitution of values with correct or consistent use of the $\frac{dy}{dx}$ value from part (b). A simplified answer is not required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

0	1	2	3 ✓

The student response accurately includes all three of the criteria below.

- ☐ implicit differentiation (2 points for full credit)
- ☐ answer

Solution:

$$\left.\frac{dy}{dx}\right|_{(x,\ y)=(1,\ 2)} = \frac{1}{2}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{6x^2-y^2}{2xy}\right) = \frac{2xy\left(12x-2y\frac{dy}{dx}\right)-\left(6x^2-y^2\right)\left(2x\frac{dy}{dx}+2y\right)}{(2xy)^2}$$

$$\left.\frac{d^2y}{dx^2}\right|_{(x,\ y)=(1,\ 2)} = \frac{4\left(12-4\cdot\frac{1}{2}\right)-(6-4)\left(2\cdot\frac{1}{2}+4\right)}{4^2} = \frac{40-10}{16} = \frac{30}{16} = \frac{15}{8}$$

3-2-2

72. If $x^2 - xy - 3y^2 = 6$, then $\frac{dy}{dx} =$

(A) $\frac{2x}{1+6y}$

(B) $\frac{2x-y}{x+6y}$

(C) $\frac{2x+y}{x+6y}$

(D) $\frac{y-2x}{x+6y}$

Answer B

Correct. The chain rule and product rule are used during the implicit differentiation to find $\frac{dy}{dx}$.

$$2x - \left(x\frac{dy}{dx} + y\right) - 6y\frac{dy}{dx} = 0 \Rightarrow 2x - y = (x + 6y)\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{2x-y}{x+6y}$$

73. $\frac{d}{dx}\left(x^5 - 5^x\right) =$

(A) $\frac{x^6}{6} - \frac{5^x}{\ln 5}$

(B) $5x^4 - 5^x$

(C) $5x^4 - x \cdot 5^{x-1}$

(D) $5x^4 - (\ln 5)\,5^x$

Answer D

Correct. The derivative of x^5 is $5x^4$ by the power rule, and the derivative of the exponential function 5^x is $(\ln 5)5^x$. Therefore, $\frac{d}{dx}\left(x^5 - 5^x\right) = 5x^4 - (\ln 5)5^x$.

3-2-2

74.

$f(1) = 4$	$f'(1) = -2$	$g(3) = 7$	$g'(3) = 1$

The point $(1, 3)$ lies on the curve in the xy-plane given by the equation $f(x)g(y) = 24 + x + y$, where f is a differentiable function of x and g is a differentiable function of y. Selected values of f, f', g, and g' are given in the table above. What is the value of $\frac{dy}{dx}$ at the point $(1, 3)$?

(A) -11

(B) 4

(C) 5

(D) 13

Answer C

Correct. The chain rule is the basis for implicit differentiation.

$f'(x)g(y) + f(x)g'(y)\frac{dy}{dx} = 1 + \frac{dy}{dx}$

$(-1 + f(x)g'(y))\frac{dy}{dx} = 1 - f'(x)g(y) \Rightarrow \frac{dy}{dx} = \frac{1-f'(x)g(y)}{-1+f(x)g'(y)}$

$\left.\frac{dy}{dx}\right|_{(1,3)} = \frac{1-f'(1)g(3)}{-1+f(1)g'(3)} = \frac{1-(-14)}{-1+4} = \frac{15}{3} = 5$

75. What is the slope of the line tangent to the curve $\sqrt{x} + \sqrt{y} = 2$ at the point $\left(\frac{9}{4}, \frac{1}{4}\right)$?

(A) -3

(B) $-\frac{1}{3}$

(C) 1

(D) $\frac{4}{3}$

Answer B

Correct. The slope of the tangent line is the value of $\frac{dy}{dx}$ at the point $\left(\frac{9}{4}, \frac{1}{4}\right)$. During the implicit differentiation to find $\frac{dy}{dx}$, both the power rule and the chain rule are needed.

$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0$

The point $\left(\frac{9}{4}, \frac{1}{4}\right)$ is on the curve, since $x = \frac{9}{4}$ and $y = \frac{1}{2}$ satisfy the equation $\sqrt{x} + \sqrt{y} = 2$. At this point,

$\frac{1}{2\left(\frac{3}{2}\right)} + \frac{1}{2\left(\frac{1}{2}\right)}\frac{dy}{dx} = 0 \Rightarrow \frac{1}{3} + \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{1}{3}.$

3-2-2

76. What is the slope of the line tangent to the curve $x^2y + y^2 = 21$ at the point $(2, 3)$?

(A) $-\frac{9}{2}$

(B) -2

(C) $-\frac{6}{5}$ ✓

(D) $\frac{9}{10}$

Answer C

Correct. Both the product rule and the chain rule are used during the implicit differentiation to find the value of $\frac{dy}{dx}$ at the point $(2, 3)$.

$2xy + x^2\frac{dy}{dx} + 2y\frac{dy}{dx} = 0$

$\left(x^2 + 2y\right)\frac{dy}{dx} = -2xy \Rightarrow \frac{dy}{dx} = -\frac{2xy}{x^2+2y}$

$\left.\frac{dy}{dx}\right|_{(2,3)} = -\frac{2(2)(3)}{2^2+2(3)} = -\frac{12}{10} = -\frac{6}{5}$

77. If $x \cos y = \pi - 2y$, then $\frac{dy}{dx} =$

(A) $\dfrac{\cos y}{x \sin y - 2}$ ✓

(B) $\dfrac{\cos y}{2 + x \sin y}$

(C) $\dfrac{-\cos y}{2 + x \sin y}$

(D) $\dfrac{x \sin y - \cos y}{2}$

Answer A

Correct. Both the product rule and the chain rule are used during the implicit differentiation to find the value of $\frac{dy}{dx}$.

$\cos y - x \sin y\frac{dy}{dx} = -2\frac{dy}{dx}$

$\cos y = -2\frac{dy}{dx} + x \sin y\frac{dy}{dx} = (x \sin y - 2)\frac{dy}{dx}$

$\frac{dy}{dx} = \frac{\cos y}{x \sin y - 2}$

3-2-2

78. If $x^3 - 2xy + 3y^2 = 7$, then $\frac{dy}{dx} =$

(A) $\frac{3x^2+4y}{2x}$

(B) $\frac{3x^2-2y}{2x-6y}$

(C) $\frac{3x^2}{2x-6y}$

(D) $\frac{3x^2}{2-6y}$

Answer B

79. If $x = \frac{1}{y+1}$, then $\frac{dy}{dx} =$

(A) $-(y+1)^2$

(B) $\frac{-1}{(y+1)^2}$

(C) $\frac{1}{(y+1)^2}$

(D) $\frac{y}{(y+1)^2}$

Answer A

Correct. During the implicit differentiation to find $\frac{dy}{dx}$, both the power rule and the chain rule are needed.

$1 = \frac{-1}{(y+1)^2}\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -(y+1)^2$

3-2-2

80. If $y = \ln(2x^2 - 3y^2)$, then $\frac{dy}{dx} =$

(A) $\frac{1}{2x^2-3y^2}$

(B) $\frac{4x}{2x^2-3y^2}$

(C) $\frac{4x-6y}{2x^2-3y^2}$

(D) $\frac{4x}{2x^2-3y^2+6y}$

Answer D

Correct. The chain rule is the basis for implicit differentiation as well as the differentiation of a composite function.
$\frac{dy}{dx} = \frac{1}{2x^2-3y^2} \cdot \left(4x - 6y\frac{dy}{dx}\right) \Rightarrow (2x^2 - 3y^2)\frac{dy}{dx} = 4x - 6y\frac{dy}{dx} \Rightarrow \left(2x^2 - 3y^2 + 6y\right)\frac{dy}{dx} = 4x \Rightarrow \frac{dy}{dx} = \frac{4x}{2x^2-3y^2+6y}$

81. If $e^{2y} - e^{(y^2-y)} = x^4 - x^2$, then the value of $\frac{dy}{dx}$ at the point $(1, 0)$ is

(A) 0

(B) $\frac{1}{2}$

(C) $\frac{2}{3}$

(D) 2

Answer C

Correct. The point $(1, 0)$ is on the curve because $x = 1$, $y = 0$ satisfies the implicit equation of the curve. The chain rule is the basis for implicit differentiation.
$2e^{2y}\frac{dy}{dx} - (2y-1)e^{(y^2-y)}\frac{dy}{dx} = 4x^3 - 2x \Rightarrow \left(2e^{2y} - (2y-1)e^{(y^2-y)}\right)\frac{dy}{dx} = 4x^3 - 2x \Rightarrow \frac{dy}{dx} = \frac{4x^3-2x}{2e^{2y}-(2y-1)e^{(y^2-y)}}$
$\left.\frac{dy}{dx}\right|_{(1,0)} = \left.\frac{4x^3-2x}{2e^{2y}-(2y-1)e^{(y^2-y)}}\right|_{(1,0)} = \frac{4-2}{2-(-1)e^0} = \frac{2}{2+1} = \frac{2}{3}$

3-2-2

82. What is the slope of the line tangent to the curve $y^3 - xy^2 + x^3 = 5$ at the point $(1, 2)$?

(A) $\frac{1}{10}$

(B) $\frac{1}{8}$

(C) $\frac{5}{12}$

(D) $\frac{11}{4}$

Answer B

Correct. The point $(1, 2)$ is on the curve since $x = 1$, $y = 2$ satisfies the implicit equation of the curve. The chain rule is the basis for implicit differentiation.

$3y^2\frac{dy}{dx} - y^2 - 2xy\frac{dy}{dx} + 3x^2 = 0$

At the point $(1, 2)$, $12\frac{dy}{dx} - 4 - 4\frac{dy}{dx} + 3 = 0 \Rightarrow 8\frac{dy}{dx} - 1 = 0 \Rightarrow \frac{dy}{dx} = \frac{1}{8}$.

83. If $\sin(x + y) = 3x - 2y$, then $\frac{dy}{dx} =$

(A) $\frac{3-\cos(x+y)}{2}$

(B) $\frac{1-\cos(x+y)}{\cos(x+y)}$

(C) $\frac{3}{2+\cos(x+y)}$

(D) $\frac{3-\cos(x+y)}{2+\cos(x+y)}$

Answer D

Correct. The chain rule is the basis for implicit differentiation as well as the differentiation of a composite function.

$\cos(x + y)\left(1 + \frac{dy}{dx}\right) = 3 - 2\frac{dy}{dx}$

$\cos(x + y) + \cos(x + y)\frac{dy}{dx} = 3 - 2\frac{dy}{dx}$

$(2 + \cos(x + y))\frac{dy}{dx} = 3 - \cos(x + y)$

$\frac{dy}{dx} = \frac{3-\cos(x+y)}{2+\cos(x+y)}$

3-2-2

84.

$f(-2) = 3$	$f'(-2) = 4$	$g(4) = 5$	$g'(4) = 2$

The point $(-2, 4)$ lies on the curve in the xy-plane given by the equation $f(x)g(y) = 17 - x - y$, where f is a differentiable function of x and g is a differentiable function of y. Selected values of f, f', g, and g' are given in the table above. What is the value of $\frac{dy}{dx}$ at the point $(-2, 4)$?

(A) -27

(B) $-\frac{11}{3}$

(C) -3 ✓

(D) $-\frac{4}{7}$

Answer C

Correct. The chain rule is the basis for implicit differentiation.

$$f'(x)g(y) + f(x)g'(y)\frac{dy}{dx} = -1 - \frac{dy}{dx}$$

$$(1 + f(x)g'(y))\frac{dy}{dx} = -1 - f'(x)g(y) \Rightarrow \frac{dy}{dx} = \frac{-1-f'(x)g(y)}{1+f(x)g'(y)}$$

$$\left.\frac{dy}{dx}\right|_{(-2,4)} = \frac{-1-f'(-2)g(4)}{1+f(-2)g'(4)} = \frac{-1-20}{1+6} = \frac{-21}{7} = -3$$

85. If $2xy^2 - 3x^2y = 6x$, then $\frac{dy}{dx} =$

(A) $\frac{6}{4y-6x}$

(B) $\frac{2y^2-2xy-6}{3x^2}$

(C) $\frac{-2y^2+6xy+6}{4xy-3x^2}$ ✓

(D) $\frac{-2y^2+6xy+3x^2+6}{4xy}$

Answer C

Correct. The chain rule is the basis for implicit differentiation.

$$\frac{d}{dx}\left(2xy^2 - 3x^2y\right) = \frac{d}{dx}(6x)$$

$$2y^2 + 2x \cdot 2y\frac{dy}{dx} - 6xy - 3x^2\frac{dy}{dx} = 6$$

$$\left(4xy - 3x^2\right)\frac{dy}{dx} = -2y^2 + 6xy + 6$$

$$\frac{dy}{dx} = \frac{-2y^2+6xy+6}{4xy-3x^2}$$

3-2-2

86. If $y = \ln(3x+4y)$, then $\frac{dy}{dx} =$

(A) $\frac{1}{3x+4y}$

(B) $\frac{3}{3x+4y}$

(C) $\frac{7}{3x+4y}$

(D) $\frac{3}{3x+4y-4}$

Answer D

Correct. The chain rule is the basis for implicit differentiation as well as the differentiation of a composite function.

$\frac{dy}{dx} = \frac{1}{3x+4y} \cdot \left(3 + 4\frac{dy}{dx}\right) \Rightarrow (3x+4y)\frac{dy}{dx} = 3 + 4\frac{dy}{dx} \Rightarrow (3x+4y-4)\frac{dy}{dx} = 3$

87. If $e^y - e^{y^2} = x - x^3$, then the value of $\frac{dy}{dx}$ at the point $(0, 1)$ is

(A) $-\frac{1}{e}$

(B) $\frac{e-1}{2e}$

(C) $\frac{1+2e}{e}$

(D) undefined

Answer A

Correct. The point $(0, 1)$ is on the curve because $x = 0$, $y = 1$ satisfies the implicit equation of the curve. The chain rule is the basis for implicit differentiation.

$e^y \frac{dy}{dx} - 2ye^{y^2}\frac{dy}{dx} = 1 - 3x^2 \Rightarrow \left(e^y - 2ye^{y^2}\right)\frac{dy}{dx} = 1 - 3x^2 \Rightarrow \frac{dy}{dx} = \frac{1-3x^2}{e^y - 2ye^{y^2}}$

$\left.\frac{dy}{dx}\right|_{(0,1)} = \left.\frac{1-3x^2}{e^y - 2ye^{y^2}}\right|_{(0,1)} = \frac{1}{e-2e} = \frac{1}{-e}$

3-2-2

88. What is the slope of the line tangent to the curve $4y^2 + xy - 2x^2 = 3$ at the point $(-1, -1)$?

(A) -5

(B) $-\frac{3}{7}$

(C) $\frac{1}{4}$

(D) $\frac{1}{3}$

Answer D

Correct. The point $(-1, -1)$ is on the curve since $x = -1$, $y = -1$ satisfies the implicit equation of the curve. The chain rule is the basis for implicit differentiation.

$8y\frac{dy}{dx} + y + x\frac{dy}{dx} - 4x = 0$

.At the point $(-1, -1)$, $-8\frac{dy}{dx} - 1 - \frac{dy}{dx} + 4 = 0 \Rightarrow -9\frac{dy}{dx} + 3 = 0 \Rightarrow \frac{dy}{dx} = \frac{1}{3}$.

89. If $\cos(4x - y) = x + y$, then $\frac{dy}{dx} =$

(A) $-1 - \sin(4x - y)$

(B) $\frac{2+4\sin(4x-y)}{\sin(4x-y)}$

(C) $-\frac{1}{1+\sin(4x-y)}$

(D) $\frac{1+4\sin(4x-y)}{-1+\sin(4x-y)}$

Answer D

Correct. The chain rule is the basis for implicit differentiation as well as the differentiation of a composite function.

$-\sin(4x - y)\left(4 - \frac{dy}{dx}\right) = 1 + \frac{dy}{dx}$

$-4\sin(4x - y) + \sin(4x - y)\frac{dy}{dx} = 1 + \frac{dy}{dx}$

$(-1 + \sin(4x - y))\frac{dy}{dx} = 1 + 4\sin(4x - y)$

$\frac{dy}{dx} = \frac{1+4\sin(4x-y)}{-1+\sin(4x-y)}$

3-2-2

90. If $3x^2 + 5x^2y^2 = 2y$, then $\frac{dy}{dx} =$

(A) $\frac{-6x}{20xy-2}$

(B) $\frac{-10xy^2-6x}{10x^2y-2}$

(C) $\frac{2-6x-10xy^2}{10x^2y}$

(D) $\frac{6x+10xy^2+10x^2y}{2}$

Answer B

Correct. The chain rule is the basis for implicit differentiation.

$\frac{d}{dx}\left(3x^2 + 5x^2y^2\right) = \frac{d}{dx}(2y)$

$6x + 10xy^2 + 10x^2y\frac{dy}{dx} = 2\frac{dy}{dx}$

$\frac{dy}{dx}\left(10x^2y - 2\right) = -10xy^2 - 6x$

$\frac{dy}{dx} = \frac{-10xy^2-6x}{10x^2y-2}$

3-2-2

91. Consider the curve defined by $2y^3 + 6x^2y - 12x^2 + 6y = 1$.

(a) Show that dy/dx $= 4x - 2xy \,/\, x^2 + y^2 + 1$.

(b) Write an equation of each horizontal tangent line to the curve.

(c) The line through the origin with slope −1 is tangent to the curve at point P. Find the x– and y–coordinates of point P.

Part A

1 point is earned for implicit differentiation

1 point is earned if verifies expression for dy/dx

$$6y^2\frac{dy}{dx} + 6x^2\frac{dy}{dx} + 12xy - 24x + 6\frac{dy}{dx} = 0$$

$$\frac{dy}{dx}\left(6y^2 + 6x^2 + 6\right) = 24x - 12xy$$

$$\frac{dy}{dx} = \frac{24x - 12xy}{6x^2 + 6y^2 + 6} = \frac{4x - 2xy}{x^2 + y^2 + 1}$$

0	1	2

The student response earns all of the following points:

1 point is earned for implicit differentiation

1 point is earned if verifies expression for dy/dx

$$6y^2\frac{dy}{dx} + 6x^2\frac{dy}{dx} + 12xy - 24x + 6\frac{dy}{dx} = 0$$

$$\frac{dy}{dx}\left(6y^2 + 6x^2 + 6\right) = 24x - 12xy$$

$$\frac{dy}{dx} = \frac{24x - 12xy}{6x^2 + 6y^2 + 6} = \frac{4x - 2xy}{x^2 + y^2 + 1}$$

Part B

3-2-2

1 point is earned if sets $\frac{dy}{dx} = 0$

1 point is earned if solves $\frac{dy}{dx} = 0$

1 point is earned if uses solutions for x to find equations of horizontal tangent lines

1 point is earned if verifies which solutions for y yield equations of horizontal tangent lines

$$\frac{dy}{dx} = 0$$

$$4x - 2xy = 2x\left(2 - y\right) = 0$$

$x = 0$ or $y = 2$

When $x = 0$, $2y^3 + \ 6y \ = 1$; $y = 0.165$

There is no point on the curve with y coordinate of 2.

$y = 0.165$ is the equation of the only horizontal tangent line

0	1	2	3	4

The student response earns all of the following points:

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1 point is earned if solves $\frac{dy}{dx} = 0$

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There is no point on the curve with y coordinate of 2.

$y = 0.165$ is the equation of the only horizontal tangent line

3-2-2

Part C

1 point is earned for setting $y = -x$

1 point is earned if substitutes $y = -x$ into equation of curve

1 point is earned if solves for x and y

$y = -x$ is equation of the line.

$2(-x)^3 + 6x^2(-x) - 12x^2 + 6(-x) = 1$

$-8x^3 - 12x^2 - 6x - 1 = 0$

$x = {}^{-1}/_{2},\ y = {}^{1}/_{2}$

0	1	2	3

The student response earns all of the following points:

1 point is earned for setting $y = -x$

1 point is earned if substitutes $y = -x$ into equation of curve

1 point is earned if solves for x and y

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$2(-x)^3 + 6x^2(-x) - 12x^2 + 6(-x) = 1$

$-8x^3 - 12x^2 - 6x - 1 = 0$

$x = {}^{-1}/_{2},\ y = {}^{1}/_{2}$

3-2-2

92. Consider the curve given by $xy^2 - x^3y = 6$.

(a) Show that $\frac{dy}{dx} = \frac{3x^2y - y^2}{2xy - x^3}$.

(b) Find all points on the curve whose x-coordinate is 1, and write an equation for the tangent line at each of these points.

(c) Find the x-coordinate of each point on the curve where the tangent line is vertical.

Part A

1 point is earned for the implicit differentiation

1 point is earned for verifies expression for $\frac{dy}{dx}$

$$y^2 + 2xy\frac{dy}{dx} - 3x^2y - x^3\frac{dy}{dx} = 0$$
$$\frac{dy}{dx}\left(2xy - x^3\right) = 3x^2y - y^2$$
$$\frac{dy}{dx} = \frac{3x^2y - y^2}{2xy - x^3}$$

0	1	2

The student response earns all of the following points:

1 point is earned for the implicit differentiation

1 point is earned for verifies expression for $\frac{dy}{dx}$

$$y^2 + 2xy\frac{dy}{dx} - 3x^2y - x^3\frac{dy}{dx} = 0$$
$$\frac{dy}{dx}\left(2xy - x^3\right) = 3x^2y - y^2$$
$$\frac{dy}{dx} = \frac{3x^2y - y^2}{2xy - x^3}$$

Part B

1 point is earned for: $y^2 - y = 6$

1 point is earned for solving for y

2 points are earned for tangent lines

3-2-2

When $x = 1,\ y^2 - y = 6$

$$y^2 - y - 6 = 0$$
$$(y-3)(y+2) = 0$$
$$y = 3,\ y = -2$$

At (1, 3), $\dfrac{dy}{dx} = \dfrac{9-9}{6-1} = 0$

Tangent line equation is $y = 3$

At (1, -2), $\dfrac{dy}{dx} = \dfrac{-6-4}{-4-1} = \dfrac{-10}{-5} = 2$

Tangent line equation is $y + 2 = 2(x - 1)$

0	1	2	3	4 ✓

The student response earns all of the following points:

1 point is earned for: $y^2 - y = 6$

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When $x = 1,\ y^2 - y = 6$

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Tangent line equation is $y + 2 = 2(x - 1)$

3-2-2

Part C

1 point is earned for sets denominator of $\dfrac{dy}{dx}$ equal to 0

1 point is earned for substituting $y = \dfrac{1}{2}x^2$ or $x = \pm\sqrt{2y}$ into the equation for the curve

1 point is earned for solving for x-coordinate

Tangent line is vertical when $2xy - x^3 = 0$

$x(2y - x^2) = 0$ gives $x = 0$ or $y = \dfrac{1}{2}x^2$

There is no point on the curve with x-coordinate 0.

When $y = \frac{1}{2}x^2$, $\frac{1}{4}x^5 - \frac{1}{2}x^5 = 6$

$$-\frac{1}{4}x^5 = 6$$

$$x = \sqrt[5]{-24}$$

0	1	2	3

The student response earns all of the following points:

1 point is earned for sets denominator of $\dfrac{dy}{dx}$ equal to 0

1 point is earned for substituting $y = \dfrac{1}{2}x^2$ or $x = \pm\sqrt{2y}$ into the equation for the curve

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Tangent line is vertical when $2xy - x^3 = 0$

3-2-2

$x(2y - x^2) = 0$ gives $x = 0$ or $y = \dfrac{1}{2}x^2$

There is no point on the curve with x-coordinate 0.

When $y = \frac{1}{2}x^2$, $\frac{1}{4}x^5 - \frac{1}{2}x^5 = 6$

$$-\tfrac{1}{4}x^5 = 6$$

$$x = \sqrt[5]{-24}$$